Lesson 1

Symmetry Groups (1.2), Group Actions (1.3) & Orbits and Stabilizers (1.4).

Def symmetric group

Let Ω be a nonempty set,

$$\mathrm{Sym}(\Omega) = \text{the set of all bijections of } \Omega$$

$\mathrm{Sym}(\Omega)$ is a group with respect to composition.

$\Omega = \{1, 2, \cdots, n\} \Rightarrow \mathrm{Sym}(\Omega) = S_n$.

Def permutation group

A permutation group is a subgroup of a symmetric group.

Remark:

If Ω and Ω' have the same cardinality, then $\mathrm{Sym}(\Omega) \doteq \mathrm{Sym}(\Omega'$

Notation

S_n $\begin{pmatrix} 1 & 2 & 3 & \cdots & n \\ \alpha_1 & \alpha_2 & \alpha_3 & \cdots & \alpha_n \end{pmatrix}$

Ex $\Omega = \{0, 1, 2, 3, 4, 5, 6\}$

$\alpha \mapsto 2\alpha + 1$: $\begin{pmatrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 3 & 5 & 0 & 2 & 4 & 6 \end{pmatrix} = (013)(254)(6)$

order: $S_6 \longrightarrow (142)(356)(4123) = (1)(4356)(2)$

$= (4356)$

Def group action

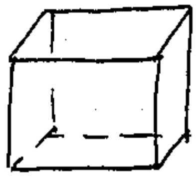
Let Ω be a nonempty set and G be a group. We say that G acts on Ω if

$G \times \Omega \to \Omega \;;\; (g, \alpha) \mapsto \alpha^g$ s.t.

1) $\alpha^e = \alpha$

2) $\alpha^{gh} = (\alpha^g)^h$

Example Symmetries of a cube



G acts on the set of vertices, of faces and of edges.

Remark:

① group action $\rightarrow$ permutation

Ω a set. G acts on Ω.

$\forall x \in G$, $x^{-1} \in G$. We have $\bar{x} : \Omega \rightarrow \Omega$; $\alpha \mapsto \alpha^{x}$

$\overline{x^{-1}} : \Omega \mapsto \Omega$; $\alpha \mapsto \alpha^{x^{-1}}$

$\Rightarrow$ $\bar{x}$ is a bijection since $\alpha^{(1)} = (\alpha)^{xx^{-1}} = (\alpha^{x})^{x^{-1}}$

② group action $\rightarrow$ permutation representation.

$\rho : G \rightarrow \mathrm{Sym}(\Omega)$; $x \mapsto \bar{x}$

since $\alpha^{hg} = (\alpha^{h})^{g}$, ρ is a group homomorphism.

ρ is called a permutation representation of G.

Def The degree of representation of an action is $|\Omega|$.

Def The kernel of a representation is $\ker\rho$.

Def A representation is faithful if $\ker\rho = 1$.

Example

① $\Omega = G$. the right regular representatio

$G \times G \rightarrow G$; $(a, x) \mapsto x^{a} = xa$ faithful.

$\Rightarrow$ every group is isomorphic to a permutation group.

② (Action on right cosets)

Let $H \leq G$ subgroup. $I_H =: \{Ha \mid a \in G\}$

G acts on I_H : $(Ha)^x = Hax$

$\rho: G \to Sym(I_H)$; $x \mapsto \bar{x}: Hg \to (Hg)^x = Hgx$

$\ker\rho = \{x \in G \mid Hgx = Hg \ \forall g \in G\}$

$\Rightarrow$ ρ is not in general faithful since $\ker\rho = \bigcap_{g\in G} g^{-1}Hg$

Def orbits and stabilizers

G acts on Ω. $\alpha \in \Omega$.

$\alpha^G = \{\alpha^g \mid g \in G\} \subset \Omega$ is the orbit of α under G.

$G_\alpha := \{g \in G \mid \alpha^g = \alpha\}$ is the stabilizer of α under G.

$\boxed{\text{Thm 1.4A}}$

G acts on Ω. $\alpha, \beta \in \Omega$. $g, h \in G$.

(1) $\alpha^G = \beta^G$ or $\alpha^G \cap \beta^G = \emptyset$

$\Rightarrow$ The set of orbits is a partition of Ω.

(2) G_α is a subgroup of G and $G_\beta = g^{-1}G_\alpha g$ if $\beta = \alpha^g$.

Moreover, $\alpha^g = \alpha^h \Rightarrow G_\alpha g = G_\alpha h$

(3) (orbit – stabilizer property) $|\alpha^G| = |G : G_\alpha|$

In particular, if G is a finite group, $|\alpha^G||G_\alpha| = |G|$

pf: (3) holds since (2) shows that the distinct points in α^G are in bijective correspondence with the right cosets of G_α in G.

Def transitive & regular

A group G acting on a set Ω is said to be transitive on Ω if it has only one orbit, and so $\alpha^G = \Omega$ for all $\alpha \in \Omega$

A group G acting transitively on a set Ω is said to act

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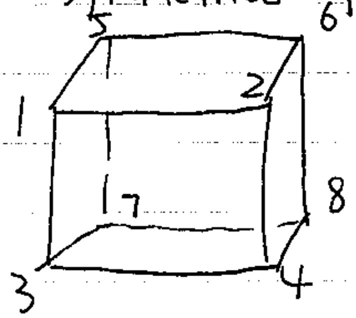
<u>regularly</u> if $G_\alpha = 1$ for each $\alpha \in \Omega$.

Corollary 1.4A

Suppose that G acts transitively on Ω then
(1) The stabilizers G_α $(\alpha \in \Omega)$ form a single conjugacy class of subgroups of G.
(2) The index $|G : G_\alpha| = |\Omega|$ for each $\alpha \in \Omega$.
(3) If G is finite, G acts regularly $\Leftrightarrow |G| = |\Omega|$

Example symmetries of cube $\Rightarrow$ calculate $|G|$

rotations & reflections



transitive :

x : rotation through an angle of $90°$ around an axis through the midpoints of the front and back faces

y : similar rotation through a vertical axis

$\Rightarrow 1^{\langle x \rangle} = \{1, 3, 4, 2\}, \quad 5^{\langle x \rangle} = \langle 5, 7, 8, 6 \rangle$

$1^{\langle y \rangle} = \{1, 2, 6, 5\}$

$\Rightarrow G \geq \langle x, y \rangle$.

① $|G : G_1| = |\Omega| = 8$

② consider the action of the subgroup G_1.

Since symmetries preserve distance, G_1 also fixes 8 and map the vertices 2, 3, 5 among themselves. ($|\text{orbit}|\max = 3$).

Now, The rotation z of $120°$ about the axis through vertice 1 and 8 induces $\bar{z} = (1)(2\,5\,3)(4\,6\,7)(8)$. So $\{2, 5, 3\}$ is indeed an orbit. So we have $|G_1 : G_{12}| = 3$.

③ Consider the action of G_{12}. G_{12} also fixes 7, 8 so G_{12} has a single nontrivial element, a reflection w in the plane through 1, 2, 7, 8 which induces $\overline{w} = (35)(46)$. Then

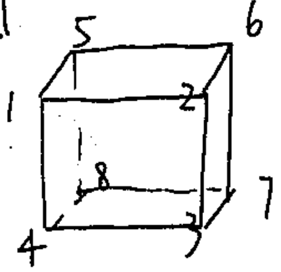
$$|G| = |G : G_1||G_1 : G_{12}||G_{12}| = 8 \cdot 3 \cdot 2 = 48.$$

Lesson 2

(1.5) (1.6)

Blocks and Primitivity, alternating group

Example



Since each symmetry preserves distances, the diagonals $\{1,7\}$, $\{5,3\}$, $\{4,6\}$, $\{2,8\}$ must be permuted amongst themselves.

i.e. G acts on the set Σ of these four pairs

If x is the rotation through 90° around the axis through the centers of the faces at the front and the back of the cube

$$\{1,7\}^x = \{2,8\}$$
$$\{2,8\}^x = \{3,5\}$$
$$\{3,5\}^x = \{4,6\}$$
$$\{4,6\}^x = \{1,7\}.$$

But this action is not faithful, since reflection in the centre of the cube leaves each of these pairs fixed.

let $\Gamma \subset \Omega$ subset. $\Gamma^g = \{\alpha^g \mid \alpha \in \Gamma\} \subset \Omega$.

G acts transitively on Ω.

Def block

We call a subset $\Delta \neq \phi \subset \Omega$ a block if for any $g \in G$, $\Delta^g = \Delta$ or $\Delta^g \cap \Delta = \phi$.

Example Symmetries of cube : acting transitively on vertices.

1) $\{1,2,3,4,5,6,7,8\}$

$\{1\},\{2\},\{3\},\{4\},\{5\},\{6\},\{7\},\{8\}$ are trivial blocks

2) $\{1,7\}$, $\{2,8\}$, $\{3,5\}$, $\{4,6\}$ are also blocks.

3) $\{1,3,6,8\}$, $\{4,2,7,5\}$ are blocks (same distance, also diagonal

In general, if G acts transitively and Δ is a block, then $\Sigma = \{\Delta^g \mid g \in G\}$ is a partition of Ω and each element of Σ is a block.

$\boxed{\text{Example}}$ $\langle (123456) \rangle$ $\Omega = \{1,2,3,4,5,6\}$

given : 5 nontrivial blocks in total $\Rightarrow$ size of blocks (2 or 3). $\begin{cases} (14)(25)(36) \\ (135)(246). \end{cases}$

Def primitive

G acts on Ω (transitively) . We say that G is primitiv[e] if there is no nontrivial blocks in Ω.

or we consider orbits

notation $\Delta \subset \Omega$. $G_{\{\Delta\}} = \{g \in G \mid \Delta^g = \Delta\}$.

$\boxed{\text{Thm 1.5A}}$

Let G acts transitively on Ω. $\alpha \in \Omega$.

Let $\mathcal{B}$ be the set of all blocks Δ for G which contains α

Let S be the set of subgroups $H \leq G$ with $G_\alpha \leq H$

Then there is a bijection between $\mathcal{B}$ and S

$\Psi : \mathcal{B} \to S$ $\Psi(\Delta) = G_{\{\Delta\}}$

$\phi : S \to \mathcal{B}$ $\phi(H) = \alpha^H = \{\alpha^h \mid h \in H\}$.

Moreover, Ψ is order-preserving : $\Delta, \Delta' \in \mathcal{B}$, $\Delta \subset \Delta' \Leftrightarrow \Psi(\Delta) \leq \Psi(\Delta')$

pf:

① Ψ, ϕ are well-defined.

$\Delta \in \mathcal{B}$. $x \in G_\alpha$ implies $\alpha \in \Delta \cap \Delta^x \Rightarrow \Delta = \Delta^x$ so $x \in G_{\{\Delta\}}$.

Suppose $\alpha^{h_1} = \alpha^{h_2 g}$, then $h_2 g h_1^{-1} \in G_\alpha \subseteq H$ so $g \in H$. so $\alpha^H \cap \alpha^{Hg} = \phi$ when $g \notin H$

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② composition is an identity.

$\psi \circ \phi$: $\alpha^{Hg} = \alpha^{H} \Leftrightarrow g \in H$

$\phi \circ \psi$: G acts transitively $\Rightarrow$ $G_{\{\Delta\}}$ transitively on Δ.

Δ is an orbit of $G_{\{\Delta\}}$

so $\alpha^{G_{\{\Delta\}}} = \Delta$

③ order preserving.

easy. $\square$.

<u>alternating group</u>

$Sym(\Omega)$ on Ω.

$FSym(\Omega)$ finitary symmetric group on Ω (permutes only f.m. point

$\Rightarrow$ $FSym(\Omega) \triangleleft Sym(\Omega)$. ($g^{-1}hg$ moves $x^g \Leftrightarrow h$ moves x, $x \in \Omega$

For any $g \in FSym(\Omega)$, g consists of m cycles of length $n_1, n_2, \cdots, n_m$ for some $m, n_1, \cdots, n_m \in \mathbb{Z}^+$

((12) : 2 cycle , $(123 \cdots k)$: k cycle) . Define

$\lambda(g) := (n_1 - 1) + \cdots + (n_m - 1) = n_1 + \cdots + n_m - m$

$supp(g) := \# \{\alpha \in \Omega \mid \alpha^g \neq \alpha\}$

If $\lambda(x)$ is even, we call x an <u>even permutation</u>

If $\lambda(x)$ is odd, we call x an <u>odd permutation</u>.

$\boxed{\text{Lemma 1-6A}}$ Sign = Sgn : $FSym(\Omega) \to \{1, -1\}$ is a group homo.

$g \mapsto (-1)^{\lambda(g)}$

need to write x as a product of disjoint cycles and so

$\lambda(x) = supp(x) - k$ where k is the number of disjoint cycles.

Lesson 3

Normal subgroups and their orbits (1.6) · Orbits and fixed points (1.7)

$\boxed{\text{Corollary of the bijection}}$ (thm 1.5A)

Let G be a group acting transitively on a set Ω with at least two points. Then

G is primitive $\Leftrightarrow$ each point stabilizer group $G_\alpha = \{g \in G \mid \alpha^g = \alpha\}$ is a maximal subgroup in G.

$\boxed{\text{Thm 1.6A}}$

Let G acts on Ω transitively, $H \lhd G$. Then

(1) any orbit of H in Ω is a block and

$\Sigma = \{\Delta^x \mid x \in G\}$ form a system of blocks

(2) if any point of Ω is fixed by all elements of H, then H lies in the kernel of the action on Ω.

($\rho: G \to \mathrm{Sym}\,\Omega$. $\ker\rho = \{g \in G \mid \alpha^g = \alpha \quad \forall \alpha \in \Omega\}$)

(3) the subgroup H has at most $|G:H|$ orbits and if $|G:H|$ is finite then the number of orbits divides $|G:H|$

(4) if G acts primitively on Ω, then either H is transitive or H lies in the kernel of the action.

pf:

(1) let Δ be an orbit of H. $\Rightarrow$ Δ^x is an orbit of $x^{-1}Hx$

(put $\beta = \alpha^{h_1 x}$, $\gamma = \alpha^{h_2 x}$, then $\gamma = \beta^{x^{-1}h_1 h_2 x}$)

Since H is normal, $x^{-1}Hx$. so Δ^x is an orbit of H

$\Rightarrow$ Since G is transitive, this means every orbit of H has the form Δ^x for some $x \in G$.

(2) by (1) all orbits of H has length 1 $\Rightarrow$ H lies in the kernel of action.

(3) this follows from (i): # of orbits $= |G : G_{\{\Delta\}}|$
Since $H < G_{\{\Delta\}}$, then $|G:G_{\alpha}| \mid |G:H|$

(4) G acts primitively $\Rightarrow$ Σ trivial $\Rightarrow$ $\Sigma = \{\Omega\}$ or $\{\{\alpha\}\}_{\alpha \in \Omega}$

<u>Def</u> fix(T) & supp(T)
Let T be a subset of G, then
$$\text{fix}(T) = \{\alpha \in \Omega \mid \alpha^g = \alpha,\ \forall g \in T\}$$
$$\text{supp}(T) = \{\alpha \in \Omega \mid \alpha^g \neq \alpha \text{ for at least one } g \in T\}$$
similarly, $\text{fix}(g) = \{\alpha \in \Omega \mid \alpha^g = \alpha\}$

<u>Orbits and fixed points</u>
Both G and Ω are finite.
$\boxed{\text{Thm } 1.7\text{A}}$ (Cauchy - Frobenius Lemma)
Let G be a finite group acting on a finite set Ω.
Suppose G has m orbits and then we have
$$\sum_{x \in G} |\text{fix}(x)| = m|G|.$$
pf: Count $|F = \{(\alpha, x) \in \Omega \times G \mid \alpha^x = \alpha\}|$ in two ways.
$$|F| = \sum_{x \in G} |\text{fix}(x)|$$
$$= \sum_{i=1}^{m} \sum_{\alpha \in \Omega_i} |G_\alpha| = \sum_{i=1}^{m} \sum_{\alpha \in \Omega_i} \frac{|G|}{|\Omega_i|} = m|G|.$$
Suppose the orbits of G are $\Omega_1, \cdots, \Omega_m$
Remark: since $|\text{fix}(x)|$ remains constant on each conjugacy class of G, we can rewrite it as $m|G| = \sum_{i=1}^{k} |C_i||\text{fix}(x_i)|$

action of G on functions

Suppose that G acts on Ω, and Γ is another set. $\mathrm{Fun}(\Omega, \Gamma)$ is the set of all functions of Δ into Γ. (e.g. elements of Γ are colours and each $\varphi \in \mathrm{Fun}(\Omega, \Gamma)$ is a colouring of the points of Ω)

If G acts on Ω, we can define action of G on $\mathrm{Fun}(\Omega, \Gamma)$:

$$\phi^x(\alpha) := \phi(\alpha^{x^{-1}}) \text{ for any } \phi \in \mathrm{Fun}(\Omega, \Gamma), x \in G, \alpha \in \Omega$$

This is indeed an action of G on $\mathrm{Fun}(\Omega, \Gamma)$:

1) $\varphi^e(\alpha) = \varphi(\alpha^e) = \varphi(\alpha)$

x^{-1} in def is to guarantee this.

2) $\forall g, h \in G$, $\varphi^{gh}(\alpha) = \varphi(\alpha^{h^{-1}g^{-1}})$

$(\varphi^g)^h(\alpha) = \varphi^g(\alpha^{h^{-1}}) = \varphi(\alpha^{h^{-1}g^{-1}})$

Corollary (from the Cauchy – Frobenius lemma)

Let Ω and Γ be finite non-empty sets and G acts on Ω. For each $x \in G$ denote by $c(x)$ the number of cycles (including 1-cycles) envolved in action of x in Ω. Then the number of orbits of G in $\mathrm{Fun}(\Omega, \Gamma)$ is

$$\frac{1}{|G|} \sum_{x \in G} |\Gamma|^{c(x)}$$

pf: $\# = \frac{1}{|G|} \sum_{x \in G} |\mathrm{fix}(x)|$ by the Cauchy - Frobenius lemma.

$\forall \varphi \in \mathrm{Fun}(\Omega, \Gamma)$, $\varphi \in \mathrm{fix}(x) \Rightarrow \varphi^x(\alpha) = \varphi(\alpha^{x^{-1}}) = \varphi(\alpha)$.

so every orbit of x is colored the same. There are $|\Gamma|^{c(x)}$ choices of such functions.

Lesson 4

Actions on k-tuples and subsets, Automorphism Groups of Algebraic structures

G acts on Ω^k, $k \geq 1$, $k \in \mathbb{Z}$: $(\alpha_1, \cdots, \alpha_k)^x = (\alpha_1^x, \alpha_2^x, \cdots, \alpha_k^x)$

easy constructions $\Omega^{(k)} \subseteq \Omega^k$: $\Omega^{(k)} = \{$k-tuples of distinct points$\}$

to generate new examples of group actions $|\Omega^{(k)}| = \binom{n}{k} \cdot k! = \frac{n!}{(n-k)!}$

$\boxed{\text{Example}}$ $\Omega = \{1, 2, 3\}$

$|\Omega^2| = |\Omega|^2$ $\Omega^2 = \{(1,2), (2,1), (1,3), (3,1), (2,3), (3,2), (1,1), (2,2), (3,3)\}$

$\Omega^{(2)} = \{(1;2), (2;1), (1;3), (3;1), (2;3), (3;2)\}$

$|\Omega^3| = |\Omega|^3$ $\Omega^{(3)} = \{(1;2;3), (1;3;2), (2;1;3), (2;3,1), (3;1;2), (3;2;1)\}$

$\boxed{\text{Example}}$ $\Omega = \{1, 2, 3, 4\}$

S_4 acts on $\Omega^{(2)}$.

orbits of (12) in $\Omega^{(2)}$: fixed points : $(3;4), (4;3)$

orbits : $(3;4)$

$(4;3)$

$(1;2), (2;1)$

$(1;3), (2;3)$

$(3;1), (3;2)$

$(1;4), (2;4)$

$(4;1), (4;2)$

Def k-transitive

We say that G is <u>k-transitive</u> if G is transitive on $\Omega^{(k)}$ where $1 \leq k \leq |\Omega|$.

We say that G is <u>highly transitive</u> if Ω is infinite and G is k-transitive for all integers $k \geq 1$.

② G acts on the set of all subsets of Ω via

$$\Gamma^x := \{\gamma^x \mid \gamma \in \Gamma\} \quad \text{for each } \Gamma \subseteq \Omega.$$

$\Rightarrow$ all subsets of a given size constitute a G-invariant set which we denote as $\Omega^{\{k\}}$ $\Rightarrow$ $|\Omega^{\{k\}}| = \binom{n}{k} = \frac{n!}{(n-k)!\,k!}$.

$\boxed{\text{Example}}$ $\Omega = \{1,2,3,4\}$

$\Omega^{\{2\}} = \{\{1,2\}, \{1,3\}, \{1,4\}, \{2,3\}, \{2,4\}, \{3,4\}\}$

<u>Def</u> k-homogeneous

G is called <u>k-homogeneous</u> if it is transitive on the set $\Omega^{\{k\}}$ $(1 \le k \le |\Omega|)$.

G is called <u>highly homogeneous</u> if Ω is infinite and G is k-homogenuous for each $k \geq 1$.

$\boxed{\text{Example}}$ $\Omega = \{1,2,3,4\}$. S_4 acts on Ω.

$H = G_{\{1,2\}} = \{g \in S_4 \mid \{1,2\}^g = \{1,2\}\}$

orbits of H in $\Omega^{\{2\}}$: $\{1,2\}$

$\{1,3\}, \{2,3\}, \{1,4\}, \{2,4\}$,

$\{3,4\}$

<u>prop</u> (ex 24 in problem set 1)

If Δ is a block and $\alpha \in \Delta$, then Δ is a union of orbits of G_α.

$\Rightarrow$ From this, if G acts inprimitively on $\Omega^{\{2\}}$, then there is a block containing $\{1,2\}$. It has to be union of some orbits of H: $\{\{1,2\}, \{3,4\}\}$ works since we have a partition

$\{\{1,2\}, \{3,4\}\}$, $\{\{2,3\}, \{1,4\}\}$, $\{\{1,3\}, \{2,4\}\}$

Take it to S_n:

$H = G_{\{1,2\}}$ .... orbits of H length

a	$\{1,2\}$		1
b	$\{1,\alpha\}, \{2,\alpha\}$	$\alpha \neq 1,2$	$2(n-2)$
c	$\{\alpha,\beta\}$	$\alpha,\beta \neq 1,2$	$\frac{(n-2)(n-3)}{2}$

$\Rightarrow n \neq 4$. $n \geq 3$, S_n acts primitively on $\Omega^{\{2\}}$:

$\{$ a∪b: $x=(1\alpha)(2\beta)$ where $\alpha \neq \beta$: $\{1,\alpha\}^x = \{1,\alpha\}$, $\{1,2\}^x = \{\alpha,\beta\}$.

a∪C: $x=(1\alpha)(2\beta)$, where $\alpha \neq \beta$: $\{\alpha,\beta\}^x = \{1,2\}$, $\{\alpha, c\}^x = \{1,c\}$ $c \neq \beta, 2$

<u>Action by automorphism group</u>

Let G be a group, an automorphism $\varphi: G \to G$ is a bijection which respects multiplication

$$\varphi(ab) = \varphi(a)\varphi(b).$$

$\Rightarrow$ $Aut(G) < Sym(G)$.

Similarly, if V is a vector space over some field F.

$Aut(V) = GL(V)$. invertible linear operators in V.

$\varphi: V \to V$ $\varphi(v_1+v_2) = \varphi(v_1)+\varphi(v_2)$

$\varphi(kv) = k\,\varphi(v)$

<u>Def</u> action preserving operation

In general, when a group acts on an algebraic structure, we shall say that the action <u>preserves</u> the structure if the elements of the group act as automorphisms.

$\boxed{\text{Example}}$ $K \trianglelefteq G$

action of G on K: $\forall g \in G$, $k \in K$, $k \mapsto k^g := g^{-1}kg$ conjugation

$\rho: G \to aut(K)$

$g \mapsto \rho(g): k \mapsto k^g$

$\ker\rho = \{g \in G \mid gk = kg \ \forall k \in K\} = C_G(K)$.

$\Rightarrow \quad G / C_G(K) < \mathrm{aut}(K)$

If $K = G$. $\quad C_G(G) = Z(G)$.

$\mathrm{Inn}(G) = \{ \varphi_g : G \to G \mid \varphi_g(h) = g^{-1}hg \} \simeq G/Z(G)$.

<u>Semidirect product</u>

① Direct product : G. H. two groups

$$G \times H = \{ (g,h) \mid g \in G,\ h \in G \}$$

multiplication defined by $(g,h) \cdot (g',h') = (gg', hh')$

② semidirect product : G. H. two groups. H acts on G respecting struct

$$G \ni g \mapsto g^x \in G, \quad x \in H.$$

Define on $G \times H$ new product :

$$(g_1, x) \cdot (g_2, y) = (g_1 g_2^{x^{-1}}, xy)$$

check : $\begin{cases} (e_G, e_H) \text{ is identity element} \\ (g, x^{-1}) = ((g^x)^{-1}, x^{-1}) \end{cases}$ & associativity

notation : $G \rtimes H$

Inside $G \rtimes H$ we have two subgroups :

$$G^* = \{ (g, e_H) \mid g \in G \} \simeq G.$$

$$H^* = \{ (e_G, x) \mid x \in H \} \simeq H$$

$G \rtimes H = G^* H^* \quad (G^* \cap H^* = \{e\})$

Moreover, G^* is a normal subgroup and H^* is acting on G^* as a conjugation :

$$\begin{aligned} (e_G, x)^{-1} (g, e_H) (e_G, x) &= (e_G, x^{-1}) (g, e_H) (e_G, x) \\ &= (g^x, x^{-1}) (e_G, x) \\ &= (g^x, e_H). \end{aligned}$$

Lesson 5

wreath (2.b) product

Introduction : an example (proved in the beginning of lesson 6).

Let Σ be a partition of a set Ω into equal-sized subsets. Then the group G of automorphisms of Σ consists of all $x \in Sym(\Omega)$ s.t. if $\Delta \subseteq \Omega$ then

$$\Delta \in \Sigma \iff \Delta^x \in \Sigma$$

Clearly, G also acts on Σ. If we define B to be the kernel of this latter action, then it is easy to see that B is isomorphic to the direct product of $|\Sigma|$ copies of $Sym(\Delta)$ where $\Delta \in \Sigma$, and that $G \cong B \rtimes Sym(\Sigma)$ where $Sym(\Sigma)$ acts on B by permuting the components of the elements of B in a natural way.

Def wreath product

If Γ and Δ are nonempty sets then we write $Fun(\Gamma, \Delta)$ to denote the set of all functions from Γ into Δ. In the case that K is a group, we can turn $Fun(\Gamma, K)$ into a group by defining a product "pointwise":

$$(fg)(\gamma) := f(\gamma)g(\gamma) \quad \text{for all } f,g \in Fun(\Gamma, K) \text{ and } \gamma \in \Gamma$$

In the case that Γ is finite of size m, $Fun(\Gamma, K)$ is isomorphic to K^m (a direct product of m copies of K).

Let H and K be groups and suppose H acts on the nonempty set Γ. Then the wreath product of K by H with respect to this action is defined to be $Fun(\Gamma, K) \rtimes H$ where H acts on $Fun(\Gamma, K)$ via

$$f^x(\gamma) = f(\gamma^{x^{-1}}) \quad \text{for all } f \in Fun(\Gamma, K), \gamma \in \Gamma, x \in H$$

We denote this group by $K \,wr_{\Gamma}\, H$.

☆ $B := \{(f, e_H) \mid f \in \mathrm{Fun}(\mathcal{I}, K)\} \cong \mathrm{Fun}(\mathcal{I}, K)$ is called the base group of the wreath product.

☆ In the special case of a wreath product where H acts regularly on itself, we write $K \text{ wr } H$ in place of $K \text{ wr}_H H$. this is called the standard wreath product.

$\boxed{\text{Example}}$

If $\mathcal{I}$ is finite, say, $\mathcal{I} = \{1, 2, 3, \cdots, m\}$.
Then action of H on B corresponds to permuting components of K^m :

$$(u_1, u_2, \cdots, u_m)^x = (u_{1'}, \cdots, u_{m'})$$

where $x = \begin{pmatrix} 1 & 2 & \cdots & m \\ 1' & 2' & \cdots & m' \end{pmatrix}$

$\boxed{\text{Thm 2.6 A}}$ (Universal embedding theorem).

Let G be an arbitrary group with a normal subgroup N, and put $K := G/N$.
Then there is an embedding $\phi: G \to N \text{ wr } K$ such that ϕ maps N onto $\mathrm{Im}\phi \cap B$ where B is the base group of $N \text{ wr } K$.

pf: $\varphi: G \twoheadrightarrow K$ with kernel N.
Let $T := \{t_u \mid u \in K\}$ be a set of right coset representations of N in G such that $\varphi(t_u) = u$ for each $u \in K$. If $x \in G$ then $\varphi(t_u x) = \varphi(t_u)\varphi(x) = u\varphi(x)$ and so $t_u x t^{-1}_{u\varphi(x)} \in N$.
Thus for each $x \in G$ we can define a function

$$f_x : K \to N$$
$$u \mapsto t_u x t^{-1}_{u\varphi(x)}. \quad \text{for all } u \in K.$$

and put $\phi(x) := (f_x, \varphi(x)) \in N \text{ wr } K$.
check :
① homomorphism
$$\phi(xy) = (f_{xy}, \varphi(x)\varphi(y))$$
where $f_{xy}(u) = t_u\, xy\, t^{-1}_{u\varphi(x)\varphi(y)}$
$$= t_u x t^{-1}_{u\varphi(x)}\ t_{u\varphi(x)}\, y\, t^{-1}_{u\varphi(x)\varphi(y)}$$
$$= f_x(u)\, f_y(u\varphi(x))$$
$$= f_x f_y^{\varphi(x)}(u).$$
② injective.
$$\begin{cases} f_x(u)=1 \Rightarrow f_x(N) = t_N x t^{-1}_N = 1 \Rightarrow x=1 \\ \varphi(x) = N \rightarrow x \in N \end{cases}$$
③ $\phi(x)$ lies in B when $\varphi(x) = N \Leftrightarrow x \in N$. □

Construct new action.

Let H and K be two groups acting on Γ and Δ respectively.
$\mathrm{Fun}(\Gamma, K)$ if Γ is finite $K^{|\Gamma|}$ copies
$\mathrm{Fun}(\Gamma, \Delta)$ if $\Delta^{|\Gamma|}$ copies
We construct action of $W := K \text{ Wr}_\Gamma H$ on $\Omega = \mathrm{Fun}(\Gamma, \Delta)$:
For each $\phi \in \Omega$ and $(f, x) \in W$:
$$\phi^{(f,x)}(\gamma) = \phi(\gamma^{x^{-1}})^{f(\gamma^{x^{-1}})}, \quad \gamma \in \Gamma.$$
check this is indeed an action:
① $(id, 1_H)$: $\phi^{(id, 1_H)}(\gamma) = \phi(\gamma^{1_H^{-1}})^{id} = \phi(\gamma)$
② $\phi^{(f,x)(g,y)}(\gamma) = \phi^{(fg^{x^{-1}}, xy)}(\gamma) \quad \forall \phi \in \Omega$.
$\phi^{(f,x)(g,y)}(\gamma^{xy}) = \phi(\gamma)^{fg^{x^{-1}}(\gamma)} = \phi(\gamma)^{f(\gamma)g(\gamma^x)}$.
$$\phi^{(f,x)(g,y)}(\gamma^{xy}) = \phi^{(f,x)}(\gamma^{xyy^{-1}})^{g(\gamma^{xyy^{-1}})}$$
$$= \phi^{(f,x)}(\gamma^x)^{g(\gamma^x)}$$
$$= \phi(\gamma)^{f(\gamma)g(\gamma^x)}.$$

Lesson 6&7 (2.7)

Primitive Wreath Products

Exercise 2.6.2

Let $G \leq Sym(\Omega)$ be an imprimitive group and let $\Sigma = \{\Gamma_i \mid i \in I\}$ be a system of blocks for G. Let H denote the kernel of the action of G on Σ and let K be the subgroup of $Sym(\Omega)$ consisting of all $x \in Sym\,\Omega$ such that $\Gamma_i^x \in \Sigma$ for each $i \in I$. Show that $K \cong Sym(\Gamma)\ wr_I\ Sym\ I$ where $|\Gamma| = |\Gamma_i|$ for each $i \in I$ and so G can be embedded in $Sym(\Gamma)\ wr_I\ Sym(I)$ in such a way that H consists of the set of elements of G which are mapped into the base group.

$\Rightarrow$ main part: $K \cong Sym(\Gamma)\ wr_I\ Sym\ I$.

proof:

① we define a map from K to $Sym(\Gamma)\ wr_I\ Sym\ I$.

$$Sym(\Gamma)\ wr_I\ Sym\ I = Fun(I, Sym(\Gamma)) \rtimes Sym\ I$$

In $Fun(I, Sym(\Gamma))$, we identify Γ_I with Γ_1. This is done by choosing any $h_1, h_2, \cdots, h_m \in K$ s.t. $h_i(\Gamma_1) = \Gamma_i$ where $|I| = m$

In this way, we have $\lambda : \Omega \to \Gamma_1 \times I$ bijection.

$$\alpha \mapsto (\alpha^{h_i^{-1}}, i)$$

Construct $\varphi : K \longrightarrow Fun(I, Sym(\Gamma_1)) \rtimes Sym(I)$

$$g \longmapsto (f_g,\ g|_\Sigma)$$

where $f_g(i)(\alpha) = ((\alpha^{h_i})^g)^{h^{-1}_{g(i)}}$, $\alpha \in \Gamma_1$

$g|_\Sigma : \Sigma \to \Sigma \in Sym(\Sigma) = Sym(I)$

② check it is a homomorphism.

$\varphi(g_1 g_2) = (f_{g_1 g_2},\ g_1 g_2|_\Sigma)$

$g_1 g_2 \in Sym\,\Omega$ so multiplication from left to right.

$\varphi(g_1)\varphi(g_2) = (f_{g_1}, g_1|_\Sigma)(f_{g_2}, g_2|_\Sigma) = (f_{g_1} f_{g_2}^{g_1|_\Sigma^{-1}},\ g_1|_\Sigma\, g_2|_\Sigma)$

$\Rightarrow$ need to check $f_{g_1g_2} = f_{g_1} f_{g_2}^{g_1|\Sigma^{-1}}$, $f_{g_1}, f_{g_2}, f_{g_1g_2} \in \mathrm{Fun}(I, \mathrm{Sym}(\Sigma))$

$f_{g_1g_2}(i)(\alpha) = \left((\alpha^{h_i})^{g_1g_2}\right)^{h^{-1}_{g_1g_2(i)}}$

$f_{g_1} f_{g_2}^{g_1|\Sigma^{-1}}(i) = f_{g_1}(i)\, f_{g_2}^{g_1|\Sigma^{-1}}(i)$ where $f_{g_2}^{g_1|\Sigma^{-1}}(i) = f_{g_2}(g_1(i))$

$f_{g_1} f_{g_2}^{g_1|\Sigma^{-1}}(i)(\alpha) = \left(f_{g_1}(i)\right)\left(f_{g_2}^{g_1|\Sigma}\right)(\alpha)$

$= \left((\alpha^{h_i})^{g_1}\right)^{\cancel{h^{-1}_{g_1(i)} h_{g_1(i)}}\, g_2 h^{-1}_{g_2(g_1(i))}}$

$= (\alpha^{h_i})^{g_1g_2h^{-1}_{g_1g_2(i)}} = f_{g_1g_2}(i)(\alpha)$

③ check it is an iso $\Rightarrow$ easy. $\square$.

Want to build primitive groups using wreath product

$\boxed{\text{Lemma 2.7A}}$

Suppose that H and K are nontrivial groups acting on the sets Γ and Δ respectively. Then the wreath product $W := K \,\mathrm{wr}_\Gamma\, H$ is primitive in the product action on $\Omega := \mathrm{Fun}(\Gamma, \Delta)$ iff:

(i) K acts primitively but not regularly on Δ

(ii) Γ is finite and H acts transitively on Γ.

(Recall : $\phi^{(f,x)}(\gamma) := \phi(\gamma^{x^{-1}})^{f(\gamma^{x^{-1}})}$ for each $\phi \in \Omega, \gamma \in \Gamma$

Remark :

K primitive and regular $\Leftrightarrow$ K is a cyclic group of prime orde

pf :

① Recall Corollary 1.5A : A group acting transitively is primitive $\Leftrightarrow$ each point stabilizer G_α is a maximal subg

Since the point stabilizers of a transitive group are all conjugate, one of the point stabilizers is maximal $\Leftrightarrow$ all are maximal.

let B be the base group of W and put

$$H_0 := \{(1,x) \in W \mid x \in H\}$$

so W is the split extension BH_0.

Fix $\delta \in \Delta$ and define $\phi_\delta \in \Omega$ by $\phi_\delta(\gamma) := \delta \ \forall \gamma \in \Gamma$.
Then the stabilizer in W of the point ϕ_δ is

$$L := \{(f,x) \in W \mid f(\gamma) \in K_\delta \text{ for all } \gamma\}.$$

$\Rightarrow$ W is primitive $\Leftrightarrow$ W is transitive & L is a maximal subgroup.

② prove the necessity of conditions (i) and (ii).

a. If H is not transitive

Let Σ be an orbit of H in Γ, then construct

$$M := \{(f,1) \in B \mid f(\gamma) \in K_\delta \text{ for all } \gamma \in \Sigma\}.$$

M is a subgroup of B which is normalized by H and

$$L \lneq MH_0 \lneq W$$

b. If Γ is infinite

Define

$$B_0 := \{(f,1) \in B \mid f \text{ has finite support on } \Gamma\}$$

then $B_0 \lhd W$ and $L \lneq LB_0 \lneq W$

c. If K is intransitive.

Let Π be an orbit of Δ, $\delta \in \Pi$

If K is intransitive, then W is intransitive

e.g. $\varphi_\delta^{(f,x)}(\gamma) = \varphi_\delta(\gamma x^{-1})^{f(\gamma x^{-1})} = \delta^{f(\gamma x^{-1})} \in \Pi$

d. If K is transitive but imprimitive

there exists R s.t. $K_\delta < R < K$ and the subgroup

$$\{(f,x) \in W \mid f(\gamma) \in R \text{ for all } \gamma\}$$

lies strictly between L and W.

e. If K is regular

Construct

$$D := \{(f,1) \in B \mid f(\gamma) = f(\gamma') \text{ for all } \gamma, \gamma'\}$$

is normalized by H_0 and then $L \lneq DH_0 \lneq W$.

③ prove the sufficiency of conditions (i) and (ii)

First we show that W is transitive

Note that $B = \{(f,1) \mid f \in \mathrm{Fun}(\Gamma, K)\}$ is transitive :

$\forall \phi \in \mathrm{Fun}(\Gamma, \Delta)$. $\phi^{(f,1)}(\gamma) = \phi(\gamma)^{f(\gamma)}$

So ϕ can be sent to any function since K is transitive.

Hence W is transitive.

It suffices to show $L \lneq M \leq W \Rightarrow M = W$.

Since $W = BH_0$, $H_0 < L$, $W = BL$. Then we have $M = (M \cap B)L$

Therefore $M \cap B > L \cap B$ and so for some γ_0, $\exists (f,1) \in M \cap B$ with $f(\gamma_0) \notin K_\delta$. idea : K_δ maximal $\Rightarrow K = \langle K_\delta, f(\gamma_0)\rangle$ ($\Rightarrow$ any point stabilizer).

Ex a primitive group G is not regular if and only if a point stabilizer G_α equals its normalizer $N_G(G_\alpha)$. (easy).

need more condition : faithful.

Since K is primitive but not regular, $K_\delta = N_K(K_\delta)$.

So for some $u \in K_\delta$, $f(\gamma_0)^{-1} u f(\gamma_0) \notin K_\delta$.

Construct $g \in \mathrm{Fun}(\Gamma, K)$. $\begin{cases} g(\gamma_0) = u \\ g(\gamma) = id_K \quad \forall \gamma \neq \gamma_0 \end{cases}$

Then since $(f,1) \in M \cap B$, $(g,1) \in L \cap B$, $(h,1) \in M \cap B$ where

$$h = f^{-1} g f g^{-1} \Rightarrow h(\gamma_0) = f^{-1}(\gamma_0) u f(\gamma_0) u^{-1} \in K K_\delta$$
$$h(\gamma) = id_K \text{ for all } \gamma \neq \gamma_0.$$

Since K is primitive, K_δ is maximal and so $K = \langle K_\delta, h(\gamma_0)\rangle$

Therefore M contains the subgroup

$$B(\gamma_0) = \{(f,1) \in B \mid f(\gamma) = id, \ \gamma \neq \gamma_0\}.$$

Besides H_0 acts on $B(\gamma_0)$: $(1,x)^{-1} B(\gamma_0)(1,x) = B(\gamma_0 x^{-1})$

Since H is transitive on $\mathcal{I}$ we conclude that $B(\gamma) \leq M \ \forall \gamma \in \mathcal{I}$

Since $\mathcal{I}$ is finite we conclude that

$$B = \prod_{\gamma \in \mathcal{I}} B(\gamma) \leq M$$

So $M = BL = W$.

Lesson 8 & Lesson 9
(2.8)
Affine and Projective groups , graphs

<u>Def</u> : 1-dimensional affine group over a field F

Let F be a field. The set A of all permutations of F of the form $F \rightarrow F$

$$t_{\alpha,\beta} : \xi \mapsto \alpha\xi + \beta \quad (\alpha, \beta \in F,\ \alpha \neq 0)$$

constitutes a subgroup of Sym(F) which is called the <u>1-dimensional affine group over F</u> and is denoted by $AGL_1(F)$.

<u>Remark</u> when F is a finite field of order q,

$AGL_1(F) = AGL_1(q)$. $|AGL_1(q)| = q(q-1)$

$\boxed{\text{prop}}$ The set of <u>translations</u> $t_{1,\beta}$ $(\beta \in F)$ forms a transitive normal abelian subgroup T of $AGL_1(F)$.

pf:

① transitive

$\forall\, \gamma \neq \gamma' \in F, \quad \gamma = t_{1,\gamma-\gamma'}(\gamma') = \gamma' + (\gamma - \gamma')$

② normal & abelian

$$t_{c,d}(t_{a,b}(\xi)) = t_{c,d}(a\xi + b) = c(a\xi + b) + d$$
$$= \underset{\neq 0}{ca}\,\xi + cb + d$$

Then $t_{c,d}^{-1} = t_{c^{-1}, -dc^{-1}}$

So
$$t_{c,d}^{-1}\, t_{1,\beta}\, t_{c,d}(\xi) = t_{c,d}^{-1}(c\xi + c\beta + d)$$
$$= t_{c^{-1},-dc^{-1}}(c\xi + c\beta + d)$$
$$= c^{-1}(c\xi + c\beta + d) - dc^{-1}$$
$$= \xi + \beta$$

So $t_{c,d}\, t_{1,\beta} = t_{1,\beta}\, t_{c,d}$

$\boxed{\text{prop}}$ $AGL_1(F)$ is a split extension of T by an abelian subgroup
$AGL_1(F)$ is 2-transitive.

<u>Remark</u> split extension.

If Q and N are two groups, then G is an <u>extension</u> of Q by N if there is a short exact sequence

$$1 \to N \xrightarrow{\iota} G \xrightarrow{\pi} Q \to 1$$

$N \downarrow$ normal subgroup, $Q \Downarrow$ quotient group.

A <u>split extension</u> is an extension with a homomorphism $s: Q \to G$ such that $\pi \circ s = id_Q$

$\Rightarrow$ An extension is split $\Leftrightarrow$ $G = N \rtimes Q$.

pf:

① split extension.

Consider $\varphi: AGL_1(F) \twoheadrightarrow F^*$

$$t_{a,b} \mapsto a$$

$\varphi(t_{c,d}\, t_{a,b}) = \varphi(t_{ca,\, cb+d}) = ca$

$\varphi(t_{c,d})\, \varphi(t_{a,b}) = ca$

$\Rightarrow$ φ is a surjective homo and $\ker \varphi = \{t_{a,b} \mid a=1\} = T$

$AGL_1(F) = T \rtimes F^*$ with $F^* \simeq \{t_{a,0}\}$ where $t_{a,b} = t_{1,b}\, t_{a,0}$

② $AGL_1(F)$ is 2-transitive

Consider (α, α') and (β, β') where $\alpha \neq \alpha'$, $\beta \neq \beta'$

Need to find $t_{a,b}$ s.t. $t_{a,b}(\alpha) = \beta$. $t_{a,b}(\alpha') = \beta'$. Then

$$\begin{cases} a\alpha + b = \beta \\ a\alpha' + b = \beta' \end{cases} \Rightarrow \begin{cases} a = \dfrac{\beta - \beta'}{\alpha - \alpha'} \\ b = \dfrac{\alpha\beta' - \alpha'\beta}{\alpha - \alpha'} \end{cases}$$ well-defined.

Def transitive extension G of $AGL_1(F)$ in $\mathrm{Sym}(F \cup \{\infty\})$

We now adjoin a new element (which we shall denote by ∞) to F to obtain a set $\Omega := F \cup \{\infty\}$. and identify $\mathrm{Sym}(F)$ with the stabilizer of ∞ in $\mathrm{Sym}(\Omega)$.

Then we can define a transitive extension G of $AGL_1(F)$ in $\mathrm{Sym}(\Omega)$ in the sense that G is transitive on Ω and $G_\infty = AGL_1(F)$. The group G consists of all fractional linear mappings of the form

$$t_{\alpha,\beta,\gamma,\delta} : \xi \longmapsto \frac{\alpha\xi+\beta}{\gamma\xi+\delta} \quad \text{with } \alpha,\beta,\gamma,\delta \in F, \ \alpha\delta-\beta\gamma \neq 0.$$

with the convention that $\begin{cases} t_{\alpha,\beta,\gamma,\delta}(\infty) = \alpha\gamma^{-1} \\ t_{\alpha,\beta,\gamma,\delta}(-\delta\gamma^{-1}) = \infty. \end{cases}$

Remark

The mapping $\begin{bmatrix} \alpha & \gamma \\ \beta & \delta \end{bmatrix} \mapsto t_{\alpha\beta\gamma\delta} \quad \alpha\delta-\beta\gamma \neq 0.$

defines a homomorphism from $GL_2(F)$ onto G whose kernel consists of the scalar matrices ($\alpha = \delta \neq 0$ and $\beta = \gamma = 0$)

That is, kernel $:= Z = \{ a I_2 \mid a \in F^* \}$. So $G \simeq GL_2(F)/Z$

The latter is called the projective general linear group of degree 2 over F and is denoted by $PGL_2(F)$

No.

higher dimensional affine and projective groups

▷ The higher dimensional affine and projective groups are automorphism groups of affine and projective geometry.

Def affine geometry

The affine geometry $AG_d(F)$ consists of points and affine subspaces constructed from the vector space F^d.

↓

Translates of the vector subspaces of F^d:

If S is a k-dimensional subspace of F^d, then

$$S+\beta := \{\alpha+\beta \mid \alpha \in S\}$$

is an affine subspace of dimension k for every $\beta \in F^d$.

$\boxed{\text{Example}}$ $AG_2(\mathbb{R}) = \{$all points of $\mathbb{R}^2$ and the straight lines in $\mathbb{R}$

Def automorphism of the affine space

An automorphism of the affine space $AG_d(F)$ is a permutation of the set of points which maps each affine subspace to an affine subspace (of the same dimension).

i.e. a permutation of the points that respects the affine geome

Def affine transformation

For each linear transformation $a \in GL_d(F)$ and vector $v \in F^d$, we define the affine transformation

$$t_{a,v} : F^d \to F^d ; \ u \mapsto ua + v$$

Each of these mappings $t_{a,v}$ is an automorphism of $AG_d(F)$.

<u>Def</u> affine group $AGL_d(F)$

The set of all $t_{a,v}$ ($a \in GL_d(F)$, $v \in F^d$) forms the <u>affine group</u> $AGL_d(F)$ of dimension $d \geq 1$ over F.

$$\varphi : AGL_d(F) \to GL_d(F)$$
$$t_{a,b} \mapsto a$$
$$\ker \varphi = T = \{t_{1,v} \mid v \in F^d\} \cong F^d$$

<u>prop</u>

1) $AGL_d(F)$ is a 2-transitive subgroup of $Sym(F^d)$.

2) $AGL_d(F)$ is a split extension of a regular normal subgroup $T := \{t_{1,v} \mid v \in F^d\}$ by a subgroup isomorphic to $GL_d(F)$.

<u>Further affine automorphisms</u> are derived from the automorphisms of the field F. $\forall \sigma \in Aut(F)$, there is a permutation of F^d defined by

$$t_\sigma : u \mapsto u^\sigma$$

where σ acts componentwise on the vector u.

The mappings t_σ ($\sigma \in Aut(F)$) form a subgroup of $Sym(F^d)$ isomorphic to $Aut(F)$.

<u>Def</u> affine semilinear transformations

The subgroup $\{t_\sigma\}$ together with $AGL_d(F)$ generates the group $A\Gamma L_d(F)$ of <u>affine semilinear transformations</u> which are of the form

$$t_{a,v,\sigma} : u \mapsto u^\sigma a + v$$

where $a \in GL_d(F)$, $v \in F^d$ and $\sigma \in Aut(F)$.

<u>Remark</u> when $d \geq 2$, it turns out that the group $A\Gamma L_d(F)$ is the full automorphism group of the affine geometry $AF_d(F)$

Moreover, $AGL_d(F)$ has a subgroup

$$ASL_d(F) = \{t_{a,v} \mid a \in \underset{\Updownarrow \ \det a = 1}{SL_d(F)},\ v \in F^d\}$$

called the <u>affine special linear group</u>

$\boxed{\text{prop}}$ ASL_d are 2-transitive in $Sym(F^d)$. if $d \geq 2$

<u>Def</u> projective groups

projective general linear group $PGL_d(F) = GL_d(F)/Z$

projective special linear group $PSL_d(F) = SL_d(F)/Z$

where $Z = \{\lambda I \mid \lambda \neq 0 \in F\}$

<u>Remark</u>

① when $d=2$, $PGL_d(F) \simeq$ the group of linear fractional mappin

$$PGL_2(F) \simeq \{t_{a,b,c,d}(u) = \frac{au+b}{cu+d},\ ad-cb \neq 0\}$$

② $d \geq 3$, $PGL_d(F)$ acts on the projective geometry $PG_{d-1}(F$

Consider $GL_d(F)$ acts on F^d by right multiplication with two orbits $\{0\}$ and $\Omega = F^d \setminus \{0\}$. This action is not primitive on Ω since there exists a system Λ of blocks :

$v, w \in \Omega$ lies in the same block $\Leftrightarrow$ $v = \lambda w$ for some $\lambda \neq$

i.e. a block in Λ consists of all nonzero scalar multiples of a given vector in Ω. We call this block a projective "point" and use $[\alpha_1, \cdots, \alpha_d]$ to denote the point containing $(\alpha_1, \cdots, \alpha_d$

Then, setting Λ to be the set of the points of the <u>projecti</u>

<u>geometry</u> $PG_{d-1}(F)$, $GL_d(F)$ acts on Λ naturally.
The kernel of this action is the group of scalars Z, so the image of this action on Λ is $GL_d(F)/Z = PGL_d(F)$
$\Rightarrow$ $PGL_d(F)$ acts faithfully on Λ.

So we have specified the points of the projective geometry $PG_{d-1}F$ as the 1-dimensional vector subspace of F^d. To complete the geometry, we define the projective subspaces to be the nonzero vector spaces of F^d:
If Σ is a vector subspace of F^d, it contains the 1-dimensional subspace spanned by any of its elements so Σ determines a set of projective points by containment. The projective dimension of a projective subspace Σ is defined to be one less than its vector space dimension.

<u>example</u>. Π denotes the set of 2-dimensional subspaces of F
Then elements of Π are "lines" in $PG_{d-1}(F)$.

$P \in PG_{d-1}(F)$ lines on a line $\ell \in \Pi$ if $P \subset \ell$.

<u>Def</u> automorphism group $Aut(PG_{d-1}(F))$

The <u>automorphism group</u> $Aut(PG_{d-1}(F))$ is the group of permutations of the points of $PG_{d-1}(F)$ which preserves <u>collinearity</u>.
This automorphism group acts in a natural way on the set Π of lines and also on the set of k-dimensional subspaces for any k.

a set of points in PG_{d-1} is said to be "collinear" if the points are all on the same line.

<u>Remark</u> ① $GL_d(F) < aut(PG_{d-1}(F))$ $\overset{\text{generates}}{\Longrightarrow}$ projective semilinear group $P\Gamma L_d(F)$
② field aut $6 < aut(PG_{d-1}(F))$

$P\Gamma L_d(F) = Aut(PG_{d-1}(F))$ if $d \geq$

No

Date

Graph

directed graph : digraph
nondirected graph : graph

Def graph

a graph g is (V, E) $\quad (\mathcal{G} = (V, E))$

V ↓ vertices, E → $V \times V$

$(\alpha, \beta) \in E$ joins α to β

(β is adjoint to α)

▷ finite graph # of $V < \infty$

infinite graph # of $V = \infty$

▷ (α, α) are permitted.

▷ path

① directed path from α to β of length k is sequence of vertices

$$\alpha_0 = \alpha, \alpha_1, \cdots, \alpha_k = \beta$$

with $(\alpha_i, \alpha_{i+1}) \in E \quad \forall i = 0, \cdots, k-1$

② nondirected graph. $(\alpha, \beta) \in E \Leftrightarrow (\beta, \alpha) \in E$

(α, α) is not permitted.

▷ connected

graph is connected if for any $\alpha, \beta \in G$, there exists a path from α to β.

▷ G acts on V.

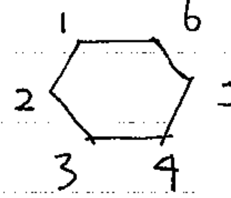
G acts on E : $(\alpha, \beta)^x = (\alpha^x, \beta^x) \quad x \in G$.

We say that G preserves the adjacency structure of g if $E^x = E$ for any $x \in G$.

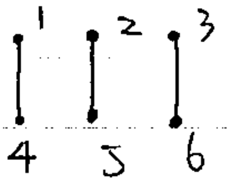
The set of all permutations of V which preserve the adjacency structure of g is called $\mathrm{aut}(\mathcal{G})$, automorphisms of $\mathcal{G}$

▷ degree of vertex : # of edges in or out.

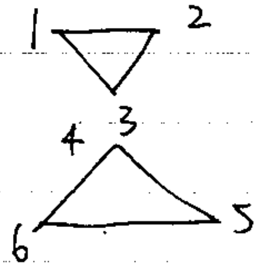
$\boxed{\text{Example}}$ aut group of $\mathcal{G}$

1) 

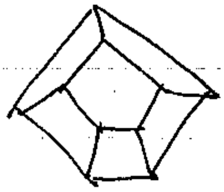
$\mathrm{aut}(g) = D_{12} = \langle (123456), (16)(25)(34) \rangle$

2) 

$\mathrm{aut}(g) = S_2 \, wr_3 \, S_3$

3) 

$\mathrm{aut}(g) = S_3 \, wr_2 \, S_2$

4) 

D_{10} & inside $\leftrightarrow$ outside

$\Downarrow$

order 20

Lesson 10

The Transitive Groups of Degree at most 7 (2.9)

Def equivalent permutation representation.

Let $\rho: G \to \mathrm{Sym}(\Omega)$ and $\sigma: G \to \mathrm{Sym}(\Gamma)$ be two permutation representation (G acts on Ω & Γ)

ρ and σ are called <u>equivalent</u> if $|\Omega| = |\Gamma|$ and there exists a bijection $\lambda: \Omega \to \Gamma$ such that

$$\lambda(\alpha^{\rho(x)}) = \lambda(\alpha)^{\sigma(x)} \quad \forall x \in G.\ \forall \alpha \in \Omega.$$

We say that two actions are equivalent when two representations are equivalent.

Observation: cases when $\Omega = \Gamma$.

If $\Omega = \Gamma$. $\lambda \in \mathrm{Sym}(\Omega)$.

$\lambda(\alpha) = \alpha^c$ for some $c \in \mathrm{Sym}(\Omega)$ $\forall \alpha \in \Omega$.

$\Rightarrow$ Two representations $\rho: G \to \mathrm{Sym}(\Omega)$, $\sigma: G \to \mathrm{Sym}(\Omega)$ are equivalent

if $\exists c \in \mathrm{Sym}(\Omega)$ s.t. $\sigma(x) = c^{-1}\rho(x)c$ $\forall x \in G$.

$\boxed{\text{prop}}$ Stabilizers in equivalent permutation representation.

Suppose G acts transitively on $\Omega^{(\rho)}$ and $\Gamma^{(\sigma)}$.

Let $\alpha \in \Omega$ and $H = G^{\rho}_{\alpha}$

Two actions are equivalent $\Leftrightarrow$ $H = G^{\sigma}_{\beta}$ for some $\beta \in \Gamma$.

pf:

$\Rightarrow$ two actions are equivalent

$\exists \lambda: \Omega \to \Gamma$ bijection and $\lambda(\alpha^{\rho(x)}) = (\lambda(\alpha))^{\sigma(x)}$.

$h \in H = G^{\rho}_{\alpha} \Rightarrow \alpha^{\rho(h)} = \alpha \Rightarrow \lambda(\alpha^{\rho(h)}) = \lambda(\alpha) = \lambda(\alpha)^{\sigma(h)}$

So $H = G^{\sigma}_{\lambda(\alpha)}$

$\Leftarrow$ $G_\alpha^\rho = H = G_\beta^\sigma$

$x \in H \iff \alpha^{\rho(x)} = \alpha$, $\beta^{\sigma(x)} = \beta$.

Define $\lambda : \Omega \to \Gamma$

$\alpha^{\rho(x)} \mapsto \beta^{\sigma(x)} \quad \forall x \in G.$

check :

① well-defined

if $\alpha^{\rho(x)} = \alpha^{\rho(y)}$, then $xy^{-1} \in H \Rightarrow \beta^{\sigma(x)} = \beta^{\sigma(y)}$

② injective

similarly as in ①

③ surjective

surjection follows from transitivity of action on Γ.

④ respects action

$\forall \gamma \in \Omega$, $\exists y$ s.t $\gamma = \alpha^{\rho(x)}$.

$\forall y \in G$, $\lambda(\gamma^{\rho(y)}) = \lambda(\alpha^{\rho(x)\rho(y)}) = \lambda(\alpha^{\rho(xy)}) = \beta^{\sigma(x)\sigma(y)}$

$= \lambda(\alpha^{\rho(x)})^{\sigma(y)} = \lambda(\gamma)^{\sigma(y)}$ $\square$

$\boxed{\text{Example}}$ G. $H \leq G$. $\Gamma_H = \{Ha \mid a \in G\}$.

Then G acts transitively on Γ_H ($|\Gamma_H| = [G:H]$) : $Ha^x = Hax$

$\rho_H : G \to \mathrm{Sym}(\Gamma_H)$ $\quad \ker \rho_H = \bigcap_{a \in G} a^{-1}Ha$

$\boxed{\text{Corollary}}$ every transitive representation of G is equivalent to ρ_H for some $H \leq G$ (since $x^{-1}Hx$ is the stabilizer of the point $Hx \in \Gamma_H$). Moreover, ρ_H and ρ_K are equivalent exactly when H and K are conjugate in G.

No.

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Example $G = S_3$

A complete set of representatives of the conjugacy classes of subgroups of G is given by 1, $\langle(12)\rangle$, $\langle(123)\rangle$, S_3. These give transitive representations of G of degree $6, 3, 2, 1$ where the first two are faithful. $|S:I|$

Moreover, if, say, S_3 acts faithfully on a set of size 8 then it must have either an orbit of size 6, or one or two orbits of size 3 with the remaining orbits being of sizes 1 or 2. (S_3 have to be faithful in one orbit)

The proper transitive groups of degree at most 7

In S_n we consider subgroups $\neq alt_n$ (proper)
$\neq S_n$

Example

① S_3 : e, $\langle(12)\rangle$, $\langle(123)\rangle$ $\Rightarrow$ no proper transitive subgr
$\langle(12)\rangle \downarrow$ not transitive, $\langle(123)\rangle \downarrow alt_3$

② S_4 : $\langle(1234)\rangle$, $\langle(12)(34), (13)(24)\rangle$, $\langle(1234), (13)\rangle$

Lemma Let $n \geq 5$. If $G \leq S_n$ and $G \neq A_n$ or S_n, then $|S_n : G| \geq n$.

pf: We use the fact that for $n \geq 5$, e, Alt_n and S_n are the only normal subgroups in S_n (will prove in chap. 3)

$G \leq S_n$, S_n acts transitively on Γ_G, $|\Gamma_G| = |S_n : G|$

$\rho_G : S_n \to Sym(\Gamma_G)$ $\ker\rho_G = \bigcap_{a \in S_n} a^{-1}Ga$

which is a maximal normal subgroup of S_n contained in G.

Since $alt_n \not\leq G$, $\ker\rho_G = \{1\}$.

So ρ_G is faithful $\Rightarrow S_n \hookrightarrow Sym(\Gamma_G)$.

Then we must have $|\Gamma_G| = |S_n : G| \geq n$.

$\boxed{n=5}$ $G \leq S_5$ transitive.

① G transitive on $\Omega = \{1,2,3,4,5\}$

$\forall G_\alpha$, $\alpha \in \Omega$, $|G : G_\alpha| = |\Omega| = 5 \Rightarrow 6 = 5n$ for some n.

② by lemma, $|S_5 : G| \geq 5 \overset{①}{\Longrightarrow} \frac{120}{5n} \geq 5 \Rightarrow n \leq 4$

③ By the Sylow theorems ($n_5 \equiv 1 \pmod 5$, $n_5 \mid |G|$),
we conclude that there is a unique (normal) Sylow 5-subgroup P of G of order 5. WLOG, take $P = \langle (12345) \rangle$

④ It suffices to find normalizers of P in S_5.
Since # of Sylow 5-subgroups in S_5 is 6, $|N_{S_5}(P)| = \frac{120}{6} = 20$

Since $\langle P, (2354) \rangle \leq N_{S_5}(P)$, $N_{S_5}(P) = \langle P, (2354) \rangle$

$\Longrightarrow$ transitive subgroup in S_5: $\langle P, (2354) \rangle$

$\langle P, (25)(34) \rangle$

P

Remark find out the structure of $\langle (12345), (2354) \rangle$

Recall $AGL_1(F) \ni t_{\alpha,\beta} : \xi \mapsto \alpha\xi + \beta$ $\alpha \neq 0$, $\alpha, \beta \in F$.

If $|F|$ is finite, then $|AGL_1(q)| = q(q-1)$ with $q = |F|$.

$AGL_1(F)$ can be generated by t_{11} and $t_{\gamma,0}$ where γ is primitive in F:

$$t_{\alpha,\beta}(\xi) = \underbrace{t_{1,1} \cdots t_{1,1}}_{\beta \text{ times}} \underbrace{t_{\gamma,0} \cdots t_{\gamma,0}}_{i \text{ times}} = \gamma^i \xi + \beta.$$

where $\gamma^i = \alpha$

where $t_{1,1} : \xi \mapsto \xi + 1 \Rightarrow (0,1,2,3,4) \mapsto (1,2,3,4,0)$

$t_{\gamma,0} : \xi \mapsto \gamma\xi \Rightarrow (0,1,2,3,4) \overset{\text{take } \gamma=2}{\longmapsto} (0,2,4,1,3)$

$\Rightarrow \langle (12345), (2354) \rangle \simeq AGL_1(5)$.

Lesson 11

The transitive Groups of Degree at most 7 Orbits of the Stabilizer

(123456) $(2,9)$ $\quad (123)(456)$ $(3,2)$ $\quad (12)(34)(56)$

$\boxed{n=6}$

① Regular groups (each permutation is of size 6)

T6.1 $\quad C_6 = \langle (123456) \rangle$

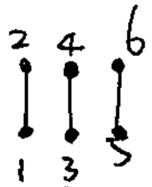
T6.2 $\quad S_3 \simeq \langle (12)(36)(45),\ (135)(246) \rangle$

② constructed by aut graphs.

If G is transitive. corresponding graph has all vertices of the same degree.

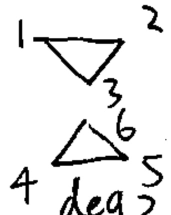
deg 2

T6.3



deg 1

T6.4



deg 2

T6.9

T6.3 $\quad D_{12} = \langle (123456),\ (16)(25)(34) \rangle$

(determined by one side) say $2 - 1$ { stays in $2 - 1$: rotation; turns to $1 - 2$: reflection.

T6.4 $\quad S_2 \,\mathrm{wr}_3\, S_3 = \langle \underbrace{(135)(246),\ (24)(13)}_{\text{generates } S_3},\ \underbrace{(12)}_{\text{generates } S_2} \rangle$

T6.9 $\quad S_3 \,\mathrm{wr}_2\, S_2$
$= \langle (14)(25)(36),\ (123),\ (12) \rangle$

Moreover : need to check transitivity.

T6.4 $\rightarrow$ T6.5 = T6.4 $\cap$ alt 6 $= \langle (135)(246),\ (24)(13),\ (12)(43) \rangle$

$\rightarrow$ T6.8 = $S_2 \,\mathrm{wr}_3\, C_3 = \langle (135)(246),\ (12) \rangle$

T6.9 $\rightarrow$ T6.10 = T6.9 $\cap$ alt 6 $= \langle (123),\ (1542)(36) \rangle$ } equal size.

$\rightarrow$ T6.11 = $\langle (123),\ (14)(25)(36),\ (12)(45) \rangle$ reflection.

$\rightarrow$ T6.12 = $C_3 \,\mathrm{wr}_2\, S_2 = \langle (14)(25)(36),\ (123) \rangle$

③ $\Omega = \{1,2,3,4\}$ $\quad S_4$ acts on $\Omega^{\{2\}} = \{\{1,2\}, \{1,3\}, \{1,4\}, \{2,3\}, \{2,4\}, \{3,4\}\}$ label by order 1,2,...

$S_4 = \langle (12), (1234) \rangle \rightarrow$ T6.6 $\simeq S_4 = \langle (24)(35),\ (1463)(25) \rangle$
$= \langle (145)(263),\ (1463)(25) \rangle$

$T6.6 \simeq S_4 = \langle (145)(263),\ (1463)(25) \rangle$

Moreover :

$T6.6 \to T6.7 = T6.6 \cap \mathrm{alt}\,6 = \langle (145)(263),\ (16)(34) \rangle$

④ primitive groups

T6.13 $\quad PGL_2(5) \qquad\qquad \alpha\delta - \gamma\beta \neq 0.$

$$GL_2(F) \to t_{\alpha\beta\gamma\delta} = \frac{\alpha\xi+\beta}{\gamma\xi+\delta} \quad \text{with } \ker = Z.$$

$$\Downarrow$$

$$F \cup \{\infty\} \to F \cup \{\infty\}.$$

$$\left\{ \begin{array}{l} |GL_2(F)/Z| = (5^2-1)(5^2-5)/4 = 120 \\ PGL_2(5) = \langle \xi \mapsto \xi+1,\ \xi \mapsto \frac{1}{\xi},\ \xi \mapsto 2\xi \rangle \end{array} \right.$$

$$= \langle (01234),\ (0\infty)(23),\ (1243) \rangle$$

T6.14 $\quad PSL_2(5) = \langle \xi \mapsto \xi+1,\ \xi \mapsto \frac{1}{\xi},\ \xi \mapsto -\xi \rangle$

$$= \langle (01234),\ (0\infty)(23),\ (14)(23) \rangle$$

Chapter 3

Primitive subgroups in S_n

n	5	6	7	8	9	…	14	16	18	20	34
$P(n)$	3	2	5	5	9		2	20	2	2	0

↳ # of proper primitive subgroups (i.e. $\neq \mathrm{alt}\,n$ or S_n)

Def proper subgroup in $\mathrm{Sym}(\Omega)$:= subgroups of $\mathrm{Sym}(\Omega)$ that does not contain $\mathrm{alt}(\Omega)$.

Def orbitals

G acts on Ω, then G acts on Ω^2 (Cartesian product of $\Omega \times \Omega$)

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where $\Omega^2 = \{(\alpha, \beta) \mid \alpha \in \Omega, \beta \in \Omega\}$. orbits of Ω^2 under G are called <u>orbitals</u>.

☆ we consider G acts transitively on Ω.

① $\Delta_1 := \{(\alpha, \alpha) \mid \alpha \in \Omega\}$ is called <u>diagonal orbital</u>

② $\Delta^* := \{(\beta, \alpha) \mid \text{for } (\alpha, \beta) \in \Delta\}$ where Δ is an orbital

Clearly $(\Delta^*)^* = \Delta$

If $\Delta^* = \Delta$, Δ is called <u>self-paired</u>. (Δ_1 is self-paired).

③ Graph (Δ) $\begin{cases} \text{Vertices} : \Omega \\ \text{edge set} : \Delta \end{cases}$

digraph (directed)

example Graph (Δ_1) :

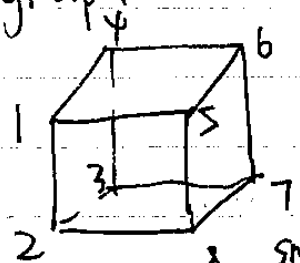
any other orbitals do not contain any loops.

<u>Remark</u> :

Since Δ is G-invariant, G acts on Graph (Δ) preserving the adjacency structure.

Example The group of symmetries of the cube acting on the set of 8 vertices. Describe orbitals and corresponding diagraph.

Graph (Δ_1) 1 2 ... 8



starting from 1 :

side (12)(14)(15)

smaller diagonal (13)(16)(18)

bigger diagonal (17)

starting from 2 :

(21)(23)(28)

(24)(25)(27)

(26).

$\Rightarrow$ three nondiagonal orbital

(3.2)

Orbits of the stabilizer

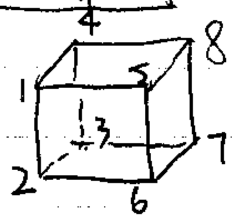
relation between orbitals and orbits of G_α

There is a relation between orbitals and orbits of G_α for $\alpha \in \Omega$.
(Since G is acting transitively all G_α are conjugated, so the choice of α is not important)

$$\text{The set of orbitals} \longrightarrow \text{The set of orbits of } G_\alpha$$
$$\Delta \longmapsto \Delta(\alpha) = \{\beta \in \Omega \mid (\alpha, \beta) \in \Delta\}$$

For if we take any (α, β), $\exists\, \Delta_0$ s.t. $(\alpha, \beta) \in \Delta_0$,
then $\forall x \in G_\alpha$, $(\alpha^x, \beta^x) \in \Delta_0$, we may consider $\{\beta \in \Omega \mid (\alpha, \beta) \in \Delta_0\}$
$\Rightarrow$ # of orbitals $=$ # of orbits of G_α $\forall \alpha \in \Omega$.
This number is called rank of G

example



Δ diagonal $\{(1,1), (2,2), \cdots\}$
Δ shortest $\{(1,2), (1,4), (1,5), (2,1), (2,3), (2,6), \cdots\}$
Δ long $\{(1,3), (1,6), (1,8), (2,4), (2,5), (2,7), \cdots\}$
Δ longest $\{(1,7), (2,8), (3,5), (4,6), (7,1), (8,2), (5,3), (6,4)\}$.

Take G_1: $\{7\}$, $\{3,6,8\}$, $\{2,4,5\}$, $\{1\}$.
$\downarrow$ Δ longest, $\downarrow$ Δ long, $\downarrow$ Δ shortest, $\downarrow$ Δ diagonal

characterize primitivity of G using Graph (Δ)

Def directed and undirected path

We say that the sequence of vertices in a graph $\mathcal{G}$ defines a directed path of length m if $v_0, v_1, \cdots, v_m$ are vertices of $\mathcal{G}$ such that $(v_i, v_{i+1}) \in \mathcal{G}$, $\forall i = 0, 1, 2, \cdots, m-1$

We call this path undirected if for $v_0, v_1, \cdots, v_m$ either $(v_i, v_{i+1}) \in \mathcal{G}$ or $(v_{i+1}, v_i) \in \mathcal{G}$

Def connected & strongly connected graph

We say that $\mathcal{G}$ is connected if for every pair of vertices u, v, there is an undirected path from u to v and $\mathcal{G}$ is strongly connected if this path can always be chosen to be directed.

$\exists$ paths of both directions

Example

$G = \langle g \rangle$ acts on $\mathbb{Z}$ by $\alpha^g = \alpha + 1$, $\forall \alpha \in \mathbb{Z}$.

Then $\Delta = \{(\alpha, \alpha+1) \mid \alpha \in \mathbb{Z}\}$ is an orbital

graph(Δ) : $\cdots \to \underset{-2}{\bullet} \to \underset{-1}{\bullet} \to \underset{0}{\bullet} \to \underset{1}{\bullet} \to \underset{2}{\bullet} \to \cdots$

connected but not strongly connected.

Thm characterization of primitivity by orbitals.

G acts transitively on Ω. Then G is primitive $\Leftrightarrow$ Graph(Δ) is connected for any non-diagonal orbital.

pf:

$(\Leftarrow)$

Suppose that Graph(Δ) is connected for any orbital Δ. Suppose Σ is some block in Ω. $|\Sigma| \geq 2$. Let $\alpha, \beta \in \Sigma$ and choose Δ such that $(\alpha, \beta) \in \Delta$, we have to show that each $\gamma \in \Omega$, $\gamma \in \Sigma$.

Since Graph(Δ) is connected, therefore $\exists$ path $v_0 = \alpha, v_1, \cdots, v_k = \gamma$. We show that $v_i \in \Sigma$ by induction on i.

$i=0$ $\alpha \in \mathcal{P}$.

Suppose $v_{i-1} \in \mathcal{P}$. $(v_{i-1}, v_i) \in \Delta$ or $(v_i, v_{i-1}) \in \Delta$.
So $\exists\, x \in G$, such that $(\alpha^x, \beta^x) = (v_{i-1}, v_i)$ or (v_i, v_{i-1})
Then α^x or β^x is equal to $v_{i-1} \in \mathcal{P} \Rightarrow \mathcal{P}^x \cap \mathcal{P} \ni v_{i-1}$
$\Rightarrow \mathcal{P}^x = \mathcal{P} \Rightarrow v_i \in \mathcal{P}. \Rightarrow \gamma \in \mathcal{P} \Rightarrow G$ is primitive.

$(\Longrightarrow)$

(Start by proving a fact:

If G is acting on Ω, we say that an equivalence relation $\approx$ on Ω is a <u>G-congruence</u> if $\alpha \approx \beta \Leftrightarrow \alpha^x \approx \beta^x$ $\forall\, x \in G$.

If G acts transitively on Ω then equivalence class of $\approx$ form a system of block for G and conversely any system of blocks provide a G-congruence equivalence on Ω.

$\Rightarrow$ Let $\mathcal{P}_\alpha$ be an equivalence class in G-congruence.

$$\mathcal{P}_\alpha := \{\beta \approx \alpha \mid \beta \in \Omega\}. \quad \mathcal{P}_\alpha{}^x = \{\beta^x \in \Omega \mid \beta \approx \alpha\}$$
$$= \mathcal{P}_{\alpha^x}$$

$\Leftarrow$ If $\mathcal{P}_i$ is a system of blocks:

$$\alpha \approx \beta \Leftrightarrow \alpha, \beta \text{ belong to the same block.}$$

easy to check $\approx$ is a G-congruence. $\square$)

Now G is primitive. Suppose Graph(Δ) is not connected for some nondiagonal orbital Δ.

Then define an equivalence relation for points in Δ:

$$\alpha \approx \beta \Leftrightarrow \alpha \text{ and } \beta \text{ are connected in Graph}(\Delta).$$

easy to check this is a G-congruence. ($\alpha \approx \alpha$ since G transitive, so $(\alpha, \beta) \in \Delta$ for some β then $\alpha \to \beta$ connects α and α.)

Thus these equivalence classes define blocks, contradiction.

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<u>Def</u> strongly primitive (as to strongly connected)

Suppose G is acting transitively on Ω. We say that G acts <u>strongly primitive</u> on Ω if Graph (Δ) is strongly connected for any non-diagonal orbital.

$\boxed{\text{Lemma}}$ conditions for primitive $\Rightarrow$ strongly primitive.

let G acts primitively on Ω

1) Graph (Δ) is strongly connected for a nondiagonal orbital Δ $\Leftrightarrow$ Graph(Δ) contains a nontrivial directed cycle

($V_0, V_1, \cdots, V_m$ directed path with $V_0 = V_m$)

2) If for each α and $\beta \in \Omega$, there exist $z \in G$ which defines cycles in Graph (Δ) containing both α and β, then G is strongly primitive.

3) If G is periodic (order of any element is finite), in particular if G is finite, then G is strongly primitive.

pf:

1)

$\Rightarrow$ trivial

$\Leftarrow$ Define an equivalence relation $\approx$ on Ω given by

$$\alpha \approx \beta \iff \alpha = \beta \text{ or } \alpha \text{ and } \beta \text{ lie on a nontrivial directed cycle in Graph}(\Delta)$$

$\approx$ is a G-congruence. Since G is primitive, the congruence classes must be either singletons or Ω itself. Since Graph(Δ) contains a nontrivial directed cycle, $\alpha \approx \beta$ for all α and β.

so Graph (Δ) is strongly connected.

2) Choose $(\alpha, \beta) \in \Delta$ and $z \in G$ such that z has a cycle of finite length containing α and β. Then $\exists$ some power y of z

s.t. $\alpha^{y} = \beta$ and $\alpha^{y^m} = \alpha$ for some $m > 1$. Thus $\alpha, \beta = \alpha^{y}, \alpha^{y^2}, \alpha^{y^3}, \cdots, \alpha^{y^m} = \alpha$ is a nontrivial directed cycle in Graph (Δ). Then Graph (Δ) is strongly connected by (i).

3) It follows immediately from 2).

The colour graph

Def binary operation $\circ$ on the set of all sets of edges of $\mathcal{G}$

Let Σ, Γ be some subsets of set of edges of $\mathcal{G}$ (i.e. $\Sigma, \Gamma \subseteq \Omega \times \Omega$). Define

$$\Sigma \circ \Gamma := \{(\alpha, \beta) \mid \exists \gamma \in \Omega \text{ s.t. } (\alpha, \gamma) \in \Sigma, (\gamma, \beta) \in \Gamma\}$$

↓

The set of all edges (α, β) in $\mathcal{G}$ such that there is a path of length 2 from α to β whose edges are from Σ and Γ respectively.

▷ The operation $\circ$ is associative $(\Sigma \circ \Gamma) \circ \Delta = \Sigma \circ (\Gamma \circ \Delta)$

> We can define "powers" by

$\Sigma^{(0)} := \Delta$, (the diagonal orbital)

$\Sigma^{(k)} := \Sigma \circ \Sigma^{(k-1)}$ for $k \geq 1$

▷ $\Sigma(\alpha) := \{\beta \in \Omega \mid (\alpha, \beta) \in \Sigma\}$ for any $\Sigma \subset \Omega \times \Omega$ and $\alpha \in \Omega$

↓

$\Sigma(\alpha)$ consists of all heads of directed edges in Σ with their tails at α.

Example

$\Sigma(1) = \{2, 3\}$.

$\boxed{\text{Thm 3.2B}}$

Let G be a group acting transitively on the set Ω.

(i) Let Σ and Λ be subsets of $\Omega\times\Omega$ and suppose that Λ is G-invariant. Then $|\Sigma\circ\Lambda(\alpha)| \le |\Sigma(\alpha)||\Lambda(\alpha)|$ for all $\alpha\in\Omega$

(ii) If G is primitive and Δ a non-diagonal orbital, put $\Gamma=\Delta\cup\Delta^*$. Then $\bigcup_{k\geq 0}\Gamma^{(k)}=\Omega\times\Omega$.

Moreover, if G has finite rank m, then $\bigcup_{k=0}^{m-1}\Gamma^{(k)}=\Omega\times\Omega$

if Graph(Δ) is strongly connected, then the same holds for Δ.

(iii) If G is primitive with finite rank r and some suborbits has finite length $m>1$, then Ω is finite and $|\Omega|\le 1+m+\cdots+m^{r-1}$.

pf:

(i) $\Sigma\circ\Lambda(\alpha) = \bigcup_{\gamma\in\Sigma(\alpha)}\Lambda(\gamma)$. Since Λ is G-invariant

$\Lambda(\alpha)^x=\Lambda(\alpha^x)$

Then $|\Lambda(\gamma)|=|\Lambda(\alpha^x)|=|\Lambda(\alpha)|$ for all $\gamma\in\Omega$.

So $|\Sigma\circ\Lambda(\alpha)| \le |\Sigma(\alpha)||\Lambda(\alpha)|$ (since we make $\Gamma=\Delta\cup\Delta^*$)

(ii) $(\alpha,\beta)\in\Gamma^{(k)} \Leftrightarrow$ there is a nondirected path of length k from α to β in Graph(Δ) so $\bigcup_{k\geq 0}\Gamma^{(k)}=\Omega\times\Omega$ follows from G primitive $\Leftrightarrow$ every nondiagonal orbital has its graph connected.

1° G has rank m

Define $\Phi(s):=\bigcup_{0\le k\le s}\Gamma^{(k)}$. We have a chain

$$\Phi(0)\subseteq\Phi(1)\subseteq\Phi(2)\subseteq\cdots$$

If $\Phi(t)=\Phi(t-1)$ for some $t\geq 1$, then $\Gamma^{(t+1)}\subseteq\Phi(t-1)\circ\Gamma\subseteq\Phi(t)$

so $\Phi(t+1) = \Phi(t)$. By induction, we have $\Phi(s) = \Phi(t-1)$ for all $s \geq t$. Moreover, since $\Delta^{(k)}$ are G-invariant, each orbital of G is contained in some $\Delta^{(k)}$. Therefore, if G has rank r, then at most r of the sets $\Phi(s)$ can differ from each other. In particular, $\Phi(r-1) = \Omega \times \Omega$.

(iii) Let Δ be an orbital for G with $|\Delta(\alpha)| = m$. Since $m > 1$, Δ is not diagonal.

① Graph (Δ) must be strongly connected.

For each $\alpha \in \Omega$, define $\Omega[\alpha]$ to be the set of all $\gamma \in \Omega$ such that there is a directed path in Graph (Δ) from α to γ.

By similar arguments in (ii), $\Omega[\alpha] = \bigcup_{0 \leq k < r} \Delta^{(k)}(\alpha)$. So $\Omega[\alpha]$ is finite by finiteness of $\Delta(\alpha)$. (they all have the same size).

By transitivity of G, $\Omega[\alpha]$ is independent of α with $|\Omega[\alpha]| \geq |\Delta(\alpha)| > 1$. Thus for any $\beta \neq \alpha$ in $\Omega[\alpha]$, we have $\Omega[\beta] = \Omega[\alpha]$ since $|\Omega[\alpha]| = |\Omega[\beta]|$ and $\Omega[\beta] \subset \Omega[\alpha]$. Thus $\alpha \in \Omega[\beta]$ and $\beta \in \Omega[\alpha]$ so there is a directed cycle in Graph (Δ) passing through α and β. Hence Graph (Δ) is strongly connected

② apply (ii) to get $\Omega[\alpha] = \Omega$ so $|\Omega| \leq \sum_{0 \leq k < r} |\Delta^{(k)}(\alpha)| \leq 1 + m + m^2 + \cdots + m^{r-1}$

<u>Def</u> subdegree

The lengths of the orbits of one of the point stabilizers are called <u>subdegrees</u>.

In the case of a finite transitive permutation of degree n, we denote subdegrees in an increasing order : $n_1 = 1, n_2, \to n_r$ where r is the rank of G.

Lesson 14

$\curlyvee$ a permutation group (i.e. a subgroup of a symmetric group)

$\boxed{\text{prop}}$ If G is primitive and not regular, then $n_i > 1$ for all $i > 1$.

pf:

① Claim: $fix(G_\alpha)$ is a block for G $\overset{\vee}{}$ if G is finite. ✓$|G|$ need finite

Indeed, $\forall \beta \in fix(G_\alpha)$, $G_\alpha \subset G_\beta \xrightarrow[\text{by transitivity}]{|G_\alpha| = |G_\beta|} G_\alpha = G_\beta$.

Suppose $\Delta = fix(G_\alpha)$.

If $\beta \in \Delta$ and $\beta^x \in \Delta$, we have $\beta^{xg} = \beta^x$ $\forall g \in G_\alpha$ which yields $xG_\alpha x^{-1} \subset G_\beta \Rightarrow xG_\alpha x^{-1} = G_\beta$.

Thus $\forall \gamma \in \Delta$, since $\gamma^{G_\beta} = \gamma$, $\gamma^{xG_\alpha x^{-1}} = \gamma$ which means $\gamma^x \in \Delta$ so $\Delta^x = \Delta$.

② If G is primitive, $fix(G_\alpha) = \{\alpha\}$ or Ω.

If $fix(G_\alpha) = \{\alpha\}$, $n_i > 1$ for all $i > 1$

If $fix(G_\alpha) = \Omega$. Then $G_\alpha = \{1\}$.

(Moreover, since G_α is maximal, G is of order p for some prime p. Since G is transitive, generator g of G is only one disjoint cycle with length $|\Omega|$ which is of order $p \Rightarrow G$ has finite prime degree.)

$\boxed{\text{prop}}$ (Lemma 3.2B (i))

Let G be a finite primitive permutation subgroup of $Sym(\Omega)$ of degree n and rank $r > 2$ with subdegrees $n_1 = 1 \le n_2 \le \cdots \le n_r$. Assume that G is not regular. Then

$$n_{i+1} \le n_i(n_2 - 1) \quad \text{for all } i \geq 2$$

(3.3)

minimal degree and bases

$\boxed{\text{Thm 3.3A}}$

let G be a primitive subgroup of $Sym(\Omega)$.

(i) If G contains a 3-cycle, then $G \geq alt(\Omega)$.

(ii) If G contains a 2-cycle, then $G \geq FSym(\Omega)$. In particular, if Ω is finite, then $G = Sym(\Omega)$.

pf:

(i) Let Δ be a maximal subset $\Delta \subset \Omega$ such that $G \geq alt(\Delta)$. Since G has a 3-cycle, ~~say $(\alpha\beta\gamma)$~~, we have $|\Delta| \geq 3$. Since G is primitive, Δ is not a block. So there exists $x \in G$ such that $\Delta \cap \Delta^x \neq \emptyset$ and $\Delta \cap \Delta^x \neq \Delta$.

1° Suppose $\Delta \cap \Delta^x$ contains just one element α. Since $x^{-1} alt(\Delta) x = alt(\Delta^x)$, $G \geq alt(\Delta)$, there are 3-cycles of the form $y := (\alpha\beta\gamma)$ and $z := (\alpha\delta\eta)$ in G where $\beta, \gamma \in \Delta$ and $\delta, \eta \in \Delta^x$. Then G contains $z^{-1}y^{-1}zy = (\alpha\beta\delta)$ with exactly two points in Δ.

2° Suppose $\Delta \cap \Delta^x$ contains at least two points α, β. Choose $\delta \in \Delta^x \setminus \Delta$. Since α, β, δ all lie in Δ^x, again G contains a 3-cycle $(\alpha\beta\delta)$ with exactly two points in Δ.

Thus $w := (\alpha\beta\delta) \in G$ with $\alpha, \beta \in \Delta$ and $\delta \notin \Delta$. Put $\Gamma :=$ $\Gamma := \Delta \cup \{\delta\}$. We claim that $G \geq alt(\Gamma)$. Since $G \geq alt(\Delta)$, it suffices to show $u \in G$ for all $u \in alt(\Gamma)$ with $\delta^u \neq \delta$. Since $\varepsilon := \delta^u \in \Delta$, there exists $v \in alt(\Delta)$ such that $\varepsilon^v = \beta$ and then $uvw \in alt(\Gamma)$ and fixes δ so uvw and vw both lie in G then $u \in G$. Hence $G \geq alt(\Gamma)$ contradicting Δ being maximal. □

(idea): put Δ maximal such that $G \geq alt(\Delta)$. Reach a contradiction by construct $|\Gamma| = |\Delta| + 1$ s.t $G \geq alt(\Gamma)$: $\delta \notin \Delta$, $\alpha, \beta \in \Delta$

primitive $\to$ Δ not a block $\to$ $\Delta \cap \Delta^x \neq \emptyset$ for some x $\to$ $\exists (\alpha\beta\delta) \in G$ $\to$ $G \geq alt(\Delta \cup \{\delta\})$

(ii) Clearly we may assume $|\Omega| \geq 3$. Suppose G contains $(\alpha\beta)$. Since $\{\alpha, \beta\}$ is not a block, $\exists\, x \in G$ such that $\{\alpha,\beta\}^x = \{\alpha,\gamma\}$ for some $\beta \neq \gamma$. Then $x^{-1}(\alpha\beta)x = (\alpha\gamma) \in G$ and $(\alpha\beta)(\alpha\gamma) = (\alpha\beta\gamma) \in G$ so by (i) $G \geq \mathrm{alt}(\Omega)$. But $\langle \mathrm{alt}(\Omega), (\alpha\beta)\rangle = \mathrm{FSym}(\Omega)$.

Def base of G

G acts on Ω.

A subset B of Ω is said to be <u>base</u> for G if

$$G_{(B)} = id$$

$\left(G_{(B)} = \bigcap_{\beta \in B} G_\beta = \{x \in G \mid \beta^x = \beta \text{ for any } \beta \in B\}.\right)$

Example

1. $G = S_n \qquad n \geq 2$

$G_{\{n\}} = S_{n-1}$

$G_{\{n\}} \cap G_{\{n-1\}} = S_{n-2}$

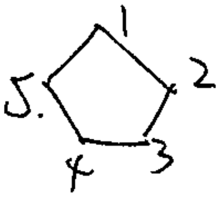
$\Rightarrow$ base : any $n-1$ points ; also n points are a base.

2. $G = A_n$

$G_{\{n\}} = A_{n-1}$

$\Rightarrow$ base : any $n-2$ points

3. $G = D_{2n}$



$G_{\{1\}} = C_2 = \{1, \rho \mid \rho^2 = 1\}$ where ρ is the reflection

$G_{\{1\}} \cap G_{\{2\}} = id$

$\Rightarrow$ base : 2 elements

No.

Date

$\boxed{\text{Exercise 3.3.2}}$ restrict $|G|$ by using |smallest base|

If G is a finite permutation group of degree n and a smallest base for G has size b, show that

$$2^b \leq |G| \leq n(n-1)\cdots(n-b+1) \leq n^b$$

pf: Denote $G_{(\ell)} = \bigcap_{j=1}^{\ell} G_{b_j}$ where $B = \{b_1, \cdots, b_b\}$.

We have

$$G \geq G_{(1)} \geq G_{(2)} \geq \cdots \geq G_{(b)} = id.$$

$$|G| = |G:G_{(1)}|\,|G_{(1)}:G_{(2)}| \cdots |G_{(b-1)}:G_{(b)}|\,\underbrace{|G_{(b)}|}_{=1}$$

1° $|G| \geq 2^b$

Since b is minimal, we must have $|G_{(i)}:G_{(i+1)}| \geq 2$, otherwise $B \setminus \{b_{i+1}\}$ is a smaller base. Thus $|G| \geq 2^b$

2° $|G| \leq n(n-1)\cdots(n-b+1)$.

Each $|G_{(i)}:G_{(i+1)}|$ is |orbit of $G_{(i)}$ containing $i+1$| $\leq n-i$

Thus $|G| \leq n(n-1)\cdots(n-b+1) \leq n^b$

<u>Def</u> minimal degree of G.

If G acts on Ω with finite support, we define the <u>minimal degree</u> of G to be the minimum of $|supp(x)|$ for all $x \in G$, $x \neq 1$.

($supp(x) = \{\alpha \in \Omega \mid \alpha^x \neq \alpha\}$).

$\boxed{\text{Exercise 3.3.1}}$ (easy). (equivalent conditions of being base)

G acts on Ω. $\Sigma \subset \Omega$. TFAE

(i) Σ is a base for G.

(ii) Σ^x is a base for G for all $x \in G$

(iii) for all $x, y \in G$, $\alpha^x = \alpha^y$ for all $\alpha \in \Sigma \Rightarrow x = y$

(iv) $\Sigma \cap supp(x) \neq \phi$ for all $x \neq 1$ in G.

Example

1) minimal degree of S_n is 2.
2) minimal degree of A_n is 3.

Exercise 1.6.7

If $x, y \in \mathrm{Sym}(\Omega)$ and $\Gamma := \mathrm{supp}(x) \cap \mathrm{supp}(y)$.
Show that $\mathrm{supp}\,[x,y] \subseteq \Gamma \cup \Gamma^x \cup \Gamma^y$. ($[x,y] := x^{-1}y^{-1}xy$)
In particular, if $|\Gamma| = 1$, show that $[x,y]$ is a 3-cycle.

pf:

① Suppose $\alpha \in \mathrm{Supp}\,[x,y]$, then it is not fixed by $x^{-1}y^{-1}xy$.

Case 1 α fixed by x and y

Impossible since otherwise it would be fixed by $x^{-1}y^{-1}xy$

Case 2 α fixed only by y.

Then $\alpha^x \neq \alpha$. $\alpha^{x^{-1}} \neq \alpha$. $\Rightarrow$ $\alpha^{x^{-1}}$ not fixed by x.

Suppose it is fixed by y. then $(\alpha^{x^{-1}})^{y^{-1}xy} = \alpha^y = \alpha$

So α is fixed by $x^{-1}y^{-1}xy$, contradiction. Thus $\alpha^{x^{-1}}$ is not fixed by x and not fixed by y, so $\alpha^{x^{-1}} \in \Gamma \Rightarrow \alpha \in \Gamma^x$

Case 3 α fixed only by x

Similarly as in Case 2, we get $\alpha \in \Gamma^y$

Case 4 α fixed by none of x and y

Then $\alpha \in \Gamma$. Then $\alpha^x \neq \alpha$. $\alpha^y \neq \alpha$. $\alpha^{x^{-1}} \neq \alpha$. $\alpha^{y^{-1}} \neq \alpha$.

② Suppose $\Gamma = \{\alpha\}$. We have $\mathrm{supp}\,[x,y] \subseteq \{\alpha, \alpha^x, \alpha^y\}$

1° $\alpha^x \neq \alpha^y$.

Otherwise $\alpha^x = \alpha^y = \beta \notin \Gamma$ so $\beta^x = \beta$ or $\beta^y = \beta \Rightarrow \alpha^y = \alpha$ or $\alpha^x = \alpha$ contradic $\wedge$

2° $\alpha^{xy} = \alpha^x$, $\alpha^{xy^{-1}} = \alpha^x$, $\alpha^{x^{-1}y} = \alpha^{x^{-1}}$, $\alpha^{x^{-1}y^{-1}} = \alpha^{x^{-1}}$

So it's easy to check $\alpha^{[x,y]} \neq \alpha$, $\alpha^{x[x,y]} \neq \alpha^x$, $\alpha^{y[x,y]} \neq \alpha^y$

$\boxed{\text{Corollary 3.3A}}$ $\mathrm{alt}\,n$ is simple for $n \geq 5$.

pf: $N \trianglelefteq \mathrm{alt}\,n$ and $N \neq 1$. We need to show $N = \mathrm{alt}\,n$.

① N is transitive

by def. no nontrivial blocks.

This is by Thm 1.6A and the primitivity of $\mathrm{alt}\,n$. $n \geq 5$

② Since $N \leq \mathrm{alt}\,n$, the minimal degree m of N is at least 3.

We show that it is in fact 3.

We may choose an element $u \in N$ of prime order, say p

so all of the nontrivial cycles of u have length p.

We observe that if $x \in \mathrm{alt}\,n$, then the commutator

$$y = [u^{-1}x^{-1}u, x] = u^{-1}(xux^{-1})u^{-1}x^{-1}ux \in N$$
$$= (u^{-1}xu)x^{-1}(u^{-1}x^{-1}u)x$$

Consider three special cases:

(i) $p > 3$, then u has a cycle $(\alpha\beta\gamma\delta\varepsilon\cdots)$ of length at least 5. Take $x = (\alpha\beta\delta)$, then $y = (\beta\delta\gamma) \in N$

(ii) $p = 3$ and u is not a 3-cycle, then u has at least two 3-cycles $(\alpha\beta\gamma)(\delta\varepsilon\theta)\cdots$. Take $x = (\alpha\beta\delta)$, then $y = (\beta\delta\varepsilon) \in N$

(iii) $p = 2$ then u has at least two cycles $(\alpha\beta)(\gamma\delta)\cdots$, Take $x = (\alpha\beta\gamma)$ then $y = (\alpha\beta)(\gamma\delta) \in N$. Take $x' = (\alpha\gamma\varepsilon)$ then $y' = (\alpha\beta\varepsilon) \in N$.

Thus N contains a 3-cycle z.

or since N contains all 3-cycles, N is primitive then by thm 3.3A, $N = \mathrm{alt}\,n$.

Since $\mathrm{alt}\,n$ is 3-transitive (in fact, $(n-2)$-transitive by taking $\sigma \in S_n$ and adjust to have $\sigma' \in \mathrm{alt}\,n$), $x^{-1}zx$ runs over the set of all 3-cycles of $\mathrm{alt}\,n$ as x runs over $\mathrm{alt}\,n$. hence N contains all 3-cycles. Then all permutations of the form $(ab)(cd)$ where a, b, c, d are distinct 4 elements belong to N since $(ab)(cd) = (abc)(adc)$. Since (abc), $(ab)(cd)$ generates $\mathrm{alt}\,n$, we have $N = \mathrm{alt}\,n$.

Lesson 15 (3.3) (3.4).

minimal degree and bases

Frobenius group it does not contain $alt(\Omega)$.

$\boxed{\text{Thm } 3.3B}$

Let $G \le Sym(\Omega)$ be a proper primitive group of finite degree n. Then G has a base of size at most $\frac{n}{2}$ and so $|G| \le n(n-1)\cdots(n-[\frac{n}{2}]+1)$.

pf: It suffices to prove the first statement. The bound on the order of G then follows from before ($2^m \le |G| \le n(n-1)\cdots(n-m+1) \le n^m$

Suppose otherwise that G has a minimal base Σ of size $> \frac{n}{2}$ elements. Then $\Omega \setminus \Sigma := \Delta$ is not a base for G since $|\Delta| \le \frac{n}{2}$. By exercise 3.3.1

Σ is a basis for G $\Leftrightarrow$ $\Sigma \cap supp(x) \ne \phi$ for all $x \ne 1$ in G

Then $\exists\, x \in G$ s.t. $\Delta \cap supp(x) = \phi$. So $supp(x) \subseteq \Omega \setminus \Delta = \Sigma$

Choose $\alpha \in Supp(x)$. Similarly $\Sigma \setminus \{\alpha\}$ is not a base so $\exists\, y \in G$ s.t. $\Sigma \setminus \{\alpha\} \cap supp(y) = \phi$. So $supp(y) \subseteq \Delta \cup \{\alpha\}$. Since Σ is a base, $\Sigma \cap supp(y) \ne \phi$. So $\alpha \in Supp(y)$ and $Supp(x) \cap Supp(y) = \{\alpha\}$ and G contains a 3-cycle $[x, y]$ by exercise 1.6.7. By thm 3.3A, $G \ge alt(\Omega)$, contradiction. $\Box$.

$\boxed{\text{Thm } 3.3C}$ restricting length of orbit. $\xrightarrow{\text{ultimately}}$ restrict $|G|$ where G transitive

Let $G \le Sym(\Omega)$ and suppose Γ and Δ are two subsets of sizes m and n respectively. If $\Gamma^x \cap \Delta \ne \phi$ for all $x \in G$ then at least one point of Γ lies in an orbit of G of length $\le mn$.

To prove this, we need the following lemma:

Let G be arbitrary group and H_i $(i=1,2,\cdots,m)$ be subgroups of G such that $G = \bigcup_{i=1}^{m} H_i x_i$ for some $x_i \in G$.

Then $|G : H_i| \le m$ for at least one i

pf of thm:

$\forall x \in G$, $\exists \gamma \in \Gamma$ s.t. $\gamma^x \in \Delta$. Denote $\delta = \gamma^x \in \Delta$. Choosing any $x_{\gamma\delta} \in G$ mapping γ to δ we have $x_{\gamma\delta} x^{-1} \in G_\gamma$ so $x \in G_\gamma x_{\gamma\delta}$. Thus every $x \in G$ lies in at least one of the cosets $G_\gamma x_{\gamma\delta}$. There are at most mn of these cosets so Lemma shows that for some $\gamma \in \Gamma$ we have $|G : G_\gamma| \le mn$, so the orbit γ^G has length $\le mn$ as required.

$\boxed{\text{Corollary 3.3B}}$

let G be a transitive subgroup of $\mathrm{Sym}(\Omega)$. If G has finite minimal degree m and a finite base of size b, then Ω is finite and $|\Omega| \le bm$

pf: let $\Delta := \mathrm{supp}(z)$ for some $z \in G$ with $|\Delta| = m$ and let Σ be a base of size b. Then $\Sigma^x \cap \Delta \neq \phi$ for all $x \in G$. Apply thm 3.3C.

Remark: Corollary 3.3B works for Ω infinite as well.

If Ω finite, there is a simple counting argument:

let Σ be a base of minimal size and let Δ be the support of an element of minimal degree. Assume $|\Omega| = n$. We have $|\Sigma^x \cap \Delta| \ge 1$ for all $x \in G$.

Count $\sum_{x \in G} |\Sigma^x \cap \Delta|$ in another way:

$\forall \alpha \in \Omega$, there are exactly $|\Sigma| G_\alpha = |\Sigma||G|/n$ values of $x \in G$ s.t. $\alpha \in \Sigma^x$. (Indeed, $\forall \beta \in \Sigma$, $\exists x$ s.t. $\alpha = \beta^x$ for $y \in G$ s.t. $\alpha = \beta^y$, $xy^{-1} \in G_\alpha$.)

Then $|\Delta||\Sigma||G|/n = \sum_{x \in G} |\Sigma^x \cap \Delta| \ge |G|$ so

$n \le bm$.

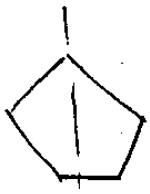
Lesson 16.
(3.4).
Frobenius group

Frobenius group

Def Frobenius group

We call a transitive permutation group (not regular) a Frobenius group if only identity element fixes more than one point. (i.e. $G_\alpha \cap G_\beta = id \quad \forall \alpha \neq \beta$).

Example

① D_{2n}, n is odd  ③ S_3 ④ A_4

② $AGL_1(F)$

$T_{\alpha\beta} : \xi \mapsto \alpha\xi + \beta \quad \alpha \neq 0, \alpha, \beta \in F.$

Suppose ξ_1, ξ_2 are fixed by $T_{\alpha\beta}$

$$\begin{cases} \xi_1 = \alpha\xi_1 + \beta \\ \xi_2 = \alpha\xi_2 + \beta \end{cases} \Rightarrow \xi_1 - \xi_2 = \alpha(\xi_1 - \xi_2) \xRightarrow{\xi_1 \neq \xi_2} \alpha = 1 \Rightarrow \beta = 0$$

Moreover, $K' = \{x \in AGL_1(F) \mid \xi^x \neq \xi \ \forall \xi \in F\} = \{T_{1\beta} \mid \beta \neq 0\}$

$K = K' \cup id \cong (F, +)$

Remark Frobenius 1901: any Frobenius group $G = K \rtimes G_\alpha$ where $K = \{id\} \cup \{x \mid \alpha^x \neq \alpha \ \forall \alpha \in \Omega\} \lhd G$.

↓

fixed point free elements.

Note that if $K < G$, then $K \lhd G$. so main difficulty is to prove K is a subgroup.

by def. $G < Sym(\Omega)$.

Observation 1 G Frobenius group and $|\Omega|$ finite $\rightarrow |G|$ finite $\rightarrow |K| = |\Omega|$

pf: $G_\alpha \cap G_\beta = 1 \quad \forall \alpha \neq \beta$

$$K = G \setminus \left(\bigcup_{\alpha \in \Omega} G_\alpha \setminus \{1\} \right)$$

$\Rightarrow |K| = |G| - |\Omega|(|G_\alpha| - 1) = |G| - |\Omega||G_\alpha| + |\Omega| = |\Omega|.$

<u>Observation 2</u> G_α acts regularly on each of its orbits on $\Omega \setminus \{\alpha\}$.
When Ω is finite, this imples $|G_\alpha|$ divides $|\Omega|-1$
($|G_\alpha| = |G_\alpha : G_{\alpha\beta}|$ for any $\beta \in \Omega \setminus \alpha$ $\Rightarrow$ each orbit has the same size).

$\boxed{\text{Thm 3.4A}}$ examine K (connecting Frobenius group and elements with order 2)

Let $G \leqslant Sym(\Omega)$ be a finite Frobenius group of degree n and let $K = \{x \in G \mid x = id \text{ or } fix(x) = \phi\}$. If G_α has even order, then K is a regular normal abelian subgroup of G and G_α has exactly one element of order 2.

pf:

Since $|G_\alpha|$ is even, it contains an element u of order 2. Let T be the G-conjugacy class containing u. Since the point stabilizers G_α, $\alpha \in G$ are conjugate and disjoint, each G_α contains at least one element from T and thus $|T| \geqslant n$. On the other hand $\forall t \in T$ has one cycle of length 1 and $\frac{n-1}{2}$ cycles of length 2 since $u \in G_\alpha$. Since no nontrivial element of G has more than one fixed point, no two elements from G can contain the same 2-cycle (otherwise $xy^{-1} \in G_{\alpha,\beta} \Rightarrow x = y$). There are exactly $\frac{n(n-1)}{2}$ 2-cycles in $Sym(\Omega)$, and so we have $|T|\frac{n-1}{2} \leq \frac{n(n-1)}{2}$ $\Rightarrow$ $|T| \leq n$ $\Rightarrow$ $|T| = n$ and each G_α contains exactly one element from T and T contains all elements of order 2 in G.

We now claim $\forall s,t \in T$, $st \in K$. Suppose $\beta^{st} = \beta$. Then $\beta^s = \beta^t$ since s,t are of order two. We then must have $\beta = \beta^s = \beta^t$ otherwise s,t contains the same cycle $(\beta\beta^t)$. But $\beta = \beta^s = \beta^t$ implies $s,t \in G_\beta$, also not possible. Thus $st \in K$ for $\forall s,t \in T$.

(observation 1)

By above discussion, $|K| = |n|$. For any $t \in T$, $Tt \subset K$. Since $|T| = n$, we conclude that $K = Tt$. In particular, $1 \in K$ and $KK^{-1} \subseteq TT \subseteq K$, ~~$KK \subseteq TT \subseteq K$~~, so K is a subgroup, then it is a normal subgroup. K is regular by def of K. Finally, since every $u \in K$ has the form $u = st$ $(s \in T)$, $t^{-1}ut = t^{-1}s = ts = u^{-1}$. Then $\forall u, v \in K$, $uv = t^{-1}(uv)^{-1}t = t^{-1}v^{-1}u^{-1}t = (t^{-1}v^{-1}t)(t^{-1}u^{-1}t) = vu$. So K is abelian. $\square$

$\boxed{\text{Thm } 3.4\text{B}}$

also known as a sharply 2-transitive group

Let $G \leq Sym(\Omega)$ be a 2-transitive Frobenius group and $K = \{x \in G \mid x = id \text{ or } fix(x) = \phi\}$. Suppose that either (i) G is finite or (ii) G_α are abelian. Then K is a regular normal abelian subgroup of G in which each nontrivial element has the same order.

pf:

~~(ii) Let T be the set of elements of order 2 in G. We claim that for any pair α, β of distinct points there is a unique $t \in T$ which maps α to β. Indeed, by 2-transitivity, $\exists t \in G$ s.t. $(\alpha, \beta)^t = (\beta, \alpha)$. Since G is a Frobenius group and t^2 fixes both α, β, $t^2 = 1 \overset{t \neq 1}{\Rightarrow} t \in T$. On the other hand, $\forall y \in T$, $\alpha^y = \beta$, $\beta^y = \alpha$ so $yt^{-1} = 1 \Rightarrow y = t$.~~

~~Next we claim that if $s, t \in T$ and neither fixes α, then $\exists x \in G_\alpha$ such that $x^{-1}sx = t$. Indeed, by 2-transitivity, we may choose $x \in G_\alpha$ such that $(\alpha, \alpha^s)^x = (\alpha, \alpha^t)$. Then $x^{-1}sx \in T$ and maps α to α^t. By uniqueness proved above, $x^{-1}sx = t$~~

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pf:

(ii) the stabilizers G_α are abelian.

Let T be the set of all elements of order 2 in G.

① $\forall$ pair of distinct points α, β, $\exists$ unique $t \in T$ s.t. $\alpha^t = \beta$.

Indeed, by 2-transitivity, $\exists\, t \in G$ such that $(\alpha, \beta)^t = (\beta, \alpha)$. Since G is a Frobenius group and t^2 fixes both α and β, $t^2 = 1$ so $t \in T$. If $s, t \in T$ s.t. $\alpha^t = \beta$, $\alpha^s = \beta$, then st^{-1} fixes α and β so $s = t$

② $\forall$ "pair of same points" α, α, $\exists$ at most one $t \in T$ s.t. $\alpha^t = \alpha$.

i.e. $\forall G_\alpha$, $\exists$ at most one element of T in G_α.

This is proved after we prove ③

③ $\forall s, t \in T$ such that neither fixes α, then $\exists\, x \in G_\alpha$ such that $x^{-1} s x = t$.

Indeed, by 2-transitivity, $\exists\, x \in G_\alpha$ such that $(\alpha, \alpha^s)^x = (\alpha, \alpha^t)$. Then $x^{-1} s x \in T$ and maps α to α^t, by ① $x^{-1} s x = t$

Now we can prove ②:

Suppose $s, t \in T \cap G_\alpha$, then by ③ $\exists\, x \in G_\beta$, $\beta \neq \alpha$ s.t. $x^{-1} s x = t$. Then $\alpha^x = \alpha^{sx} = \alpha^{xt}$ so t fixes both α^x and α yielding $\alpha^x = \alpha$. Then $x = 1$ so $s = t$.

---

① connecting K with T

$\forall 1 \neq x \in K$, $\exists\, t \in T$ such that either $x = t$ or $xt \in T$.

Indeed, by ① above, let $\beta = \alpha^x \neq \alpha$, then $\exists!\, t \in T$ such that $\beta^t = \alpha$ so $xt \in G_\alpha$. if $x \neq t$, $xt \neq 1$, $\forall \beta \neq \alpha$, xt does not fix β. Then by ① $\exists!\, s \in T$ s.t. $(\beta, \beta^{xt})^s = (\beta^{xt}, \beta)$ so $xts \in G_\beta$. If $s \in G_\alpha$, then xts fixes both α and β so $xt = s \in T$. If $s \notin G_\alpha$, by ③ $\exists\, z \in G_\alpha$ s.t. $z^{-1} s z = t$.

Since G_α is abelian, xt and z commute. Then $\beta^{xtsz} = \beta^{xtzt} = \beta^{z)}$ $= \beta^z$ thus x fixes β^z, contradiction. Therefore $s \in G_\alpha$ and $xt = s \in T$

② Suppose X is not a subgroup

Then use criterion :

K is a subgroup $\iff$ for all α, distinct elements of K lie in distinct cosets of G_α

$\exists$ distinct elements $x, y \in K$ such that $xy^{-1} \in G_\alpha$

Then with $\beta := \alpha^x = \alpha^y$ there is a unique element $t \in T$ such that $\beta^t = \alpha$. By ① either $x = y = t$ or xt and yt are both elements of $T \cap G_\alpha$ or $x = t$ any $yt \in T \cap G_\alpha$.

Neither of them is possible

(For the third case :

$\forall \beta \neq \alpha$, t and yt do not fix β.

$\exists z \in G_\beta$ s.t. $z^{-1} yt z = t \Rightarrow ytz = zt$

$\Rightarrow \alpha^{ytz} = \alpha^{zt} = \alpha^z$

$\Rightarrow t$ fixes α^z, contradiction since $x = t \in K$.

Lesson 17

The (4.3) socle

The socle

not necessary. Just the setting we would use later.

Def minimal normal subgroup

Let G be a finite permutation group (nontrivial). A minimal normal subgroup of group G is a normal subgroup K, $K \neq id$ which does not contain another normal nontrivial subgroup.

$\boxed{\text{Example}}$

- Simple group G is minimal normal subgroup of itself
- In the case of infinite cyclic group, there is no minimal normal subgroup.

Def socle

The socle of G is the subgroup generated by all minimal normal subgroups of G. denoted by $Soc(G)$.

If G does not have minimal normal subgroup, then $soc(G) := id$

$\boxed{\text{Example}}$

$G = C_{12} = \langle a \rangle$.

$\Rightarrow$ minimal normal subgroups: $\langle a^6 \rangle, \langle a^4 \rangle$

$\Rightarrow soc(G) = \langle a^2 \rangle$

Observation

{all minimal normal subgroups} is mapped to itself by autos of G (auto of G preserves normal & minimal).

$\Rightarrow Soc(G)$ is characteristic group

(i.e. it is mapped to itself by any automorphism of G).

for a finite group

Thm 4.3 A study the structure of minimal normal subgroups and $soc(G)$

Let G be a nontrivial finite group.

1) If K is minimal normal and L is some normal subgroup in G Then either $K \leq L$ or $K \cap L = id$ and $\langle K, L \rangle = K \times L = KL$

2) there exist minimal normal subgroups $K_1, \cdots, K_m$ of G such that $soc(G) = K_1 \times K_2 \times \cdots \times K_m$

3) Every minimal normal subgroup K of G is a direct product $K = T_1 \times \cdots \times T_k$ where T_k are simple normal subgroups of K which are conjugated under G.

4) If the subgroups K_i in (2) are all nonabelian, then K_i's are the only minimal normal subgroups of G. Similarly in 3) T_i's are the only minimal normal subgroups in K.

pf:

1) $K \cap L \lhd G \Rightarrow K \cap L \lhd K \Rightarrow K \cap L = K$ or id.
either $K \leq L$ or $K \cap L = id$. If $K \cap L = id$, K and L both normal, then $\forall x \in K, y \in L$, $x\underbrace{yx^{-1}y^{-1}} \in K \cap L = id \Rightarrow K$ and L commute. Then $\langle K, L \rangle = KL = K \times L$.

2). G is finite. Consider $S = \{K_1, \cdots, K_r\}$ maximal set of minimal normal subgroups K_i with respect to the property that $\langle K_1, \cdots, K_r \rangle = K_1 \times \cdots \times K_r =: H$. Want to show $H = soc(G)$. Let $K \leq G$ minimal normal, then H is normal $\Rightarrow K \leq H$ or $\langle K, H \rangle = K \times H$. The latter contradicts with maximality of H, so $K \leq H$.

3). Let T be a minimal normal subgroup of K. Then all conjuga[tes] $x^{-1}Tx$, $x \in G$ is a minimal normal subgroup of K. ($g^{-1}x^{-1}txg \in T \Rightarrow g^{-1}x^{-1}txg \in x^{-1}Tx$). Let $S = \{T_1, \cdots, T_k\}$ be maximal set of $x^{-1}Tx$, $x \in G$ with respect to the property that.

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$L = \langle T_1, \cdots, T_k \rangle = T_1 \times \cdots \times T_k$. Then L contains all $x^{-1} T x$, $x \in G$. If not, $x_1^{-1} T x_1 \cap L = id$. then $\langle x_1^{-1} T x_1, L \rangle = L \times x_1^{-1} T x_1$ contradicting maximality of S.

Thus L is normal in G. Since $L \le K$, $K = L$.

To prove T_i are simple, note that any normal subgroup T' of T_i are normal in K since T_i and T_j commute and $K = T_1 \times \cdots \times T_k$, then by minimality $T' = T_i$ or id.

4) Let K be a minimal normal subgroup of G that is distinct from the K_i ($i = 1, 2, \cdots, m$). By 1), $K \cap K_i = id$ and K commutes with each K_i so K commute with $soc(G)$ which is exactly $K \le Z(soc(G))$. For any K_i, $K_i = T_1^i \times T_2^i \times \cdots \times T_{s(i)}^i$ for some $s(i)$ where $Z(T_j^i) \triangleleft K_i$ if we notice $\forall t \in T_j^i$, $x \in Z(T_j^i)$, $k \in K_i$ $\quad k^{-1} x k t k^{-1} x^{-1} k t^{-1} = k^{-1} (k t k^{-1}) x x^{-1} k t^{-1} = 1$ since $T_j^i \triangleleft K_i$

Since T_j^i is simple, $Z(T_j^i) = id$ or T_j^i. Since T_j^i are conjugated, $Z(K^i) = K^i$ or id. Since K_i is nonabelian, we have $Z(K_i) = id$ so $k = id$ contradiction to our choice of K.

$\boxed{\text{Corollary 4.3A}}$

Every minimal normal subgroup of a finite group is either an elementary abelian p-group for some prime p or its center is equal to 1.

pf:

The latter case is proved. In the first case, each T_j^i is abelian and simple, then it must be a p-group for some prime p.

Lesson 18

(Exercise)

Let G be a nontrivial group and consider two ways in which G can act on itself:

i) (Right multiplication) $\rho : G \to Sym(G)$ defined by $a^{\rho(x)} := ax$;

ii) (Left multiplication) $\lambda : G \to Sym(G)$ defined by $a^{\lambda(x)} := x^{-1}a$

Show that the images of ρ and λ centralizes each other and that for some t of order 2 in $Sym(G)$ we have $t^{-1}\rho(G)t = \lambda(G)$.

$\Rightarrow$ $\lambda(G)$ and $\rho(G)$ are conjugated subgroups of $Sym(G)$. $t : a \mapsto a^{-1}$

1) generalize above exercise to action of G on right subsets of $H<G$ and examine the relationship between $\rho(G)$ and $\lambda(N_G(H))$.

Fix a subgroup H of the group G. Then G acts by right multiplication on the set $\mathcal{I}_H$ of right cosets of H and denote this permutation representation by ρ. Consider $K := N_G(H)$ and $\lambda(K)$ on $\mathcal{I}_H$ given by left multiplication, then $(Ha)^{\lambda(x)} = x^{-1}(Ha) = Hx^{-1}a$.

<u>Def</u> semiregularly.

A Group G acts <u>semiregularly</u> on a set Ω if G acts on Ω in such a way that $G_\alpha = 1$ $\forall \alpha \in \Omega$.

(In particular, semiregular + transitive $\Leftrightarrow$ regular).

$\boxed{\text{Lemma 4.2A}}$ need G to be finite to have $N_G(H) = \{g \mid gHg^{-1} \subseteq H\}$

Let $H < G$ and put $K := N_G(H)$. Let $\mathcal{I}_H$ denote the set of right cosets of H in G and let ρ and λ denote the right and left actions of G and K respectively on $\mathcal{I}_H$ as defined above. Then

In general not true for G infinite. $G = GL_2(\mathbb{Q})$. $N = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ $n \in \mathbb{Z}$. $g = \begin{pmatrix} 2 & \\ & 1 \end{pmatrix}$.

ex in P88 Dummit Abstract Algebra. $gNg^{-1} \subseteq N$ but $gN^{-1}g \not\subseteq N \Rightarrow g$ does not normalize N.

(i) $\ker \lambda = H$ and $\lambda(K)$ is semiregular

(ii) The centralizer C of $\rho(G)$ in $\mathrm{Sym}(\mathcal{I}_H)$ equals $\lambda(K)$.

(iii) $H \in \mathcal{I}_H$ has the same orbit under $\lambda(K)$ as under $\rho(K)$.

(iv) If $\lambda(K)$ is transitive, then $K = G$ and $\lambda(G)$ and $\rho(G)$ are conjugated in $\mathrm{Sym}(\mathcal{I}_H)$.

Pf:

(i) obviously $\ker\lambda = H$ and the point stabilizer of each point $Ha \in \mathcal{I}_H$ under λ is H so $\lambda(K) \simeq K/H$ is semiregular.

(ii) easy to see $\lambda(K) \leq C$

Conversely, we have $\forall \varphi \in C$, $\rho_a \varphi = \varphi \rho_a$ $\forall a \in G$.

Then $\varphi(Hxa) = \varphi(Hx)a$ $\forall x, a \in G$.

In particular $\varphi(Ha) = \varphi(H)a$ $\forall a \in G$.

Suppose $\varphi(H) = Hb$ Then $\varphi(Ha) = Hba$. $\forall a \in G$.

In particular let $a \in H$, Then $Hb = Hba \Rightarrow bab^{-1} \in H \Rightarrow bHb^{-1} \subseteq H$ $\Rightarrow b \in K$. so $\varphi = \lambda(b^{-1}) \in \lambda(K)$.

(iii). In both cases the orbit of H is the set of right cosets of H in K.

(iv). If $\lambda(K)$ is transitive, $\forall g \in G$, $\exists k \in K$ s.t. $Hg = Hk \Rightarrow gk^{-1} \in H \Rightarrow g \in Hk \subseteq K$ so $G = K = N_G(H)$ and $H \lhd G$.

We can define $t \in \mathrm{Sym}(\mathcal{I}_H)$ by $(Ha)^t := Ha^{-1}$; when $H \lhd G$ this mapping is defined (indeed, if $Ha = Hb$, then $ab^{-1} \in H$, then $b^{-1}(ab^{-1})b \in H$ by normality so $b^{-1}a \in H$, so $Hb^{-1} = Ha^{-1}$).

It is easy to verify $t^{-1}\lambda(x)t = \rho(x)$ $\forall x \in G$.

Def permutation isomorphic
G acts on Δ and H acts on Γ. If $\exists$ iso $\varphi : G \to H$ and bijection $\psi : \Delta \to \Gamma$ s.t. $\psi(\alpha^g) = \psi(\alpha)^{\varphi(g)}$ $\forall \alpha \in \Delta$, $g \in G$ then the two group actions are said to be <u>permutation isomorphic</u>

Lesson 19

$\boxed{\text{Thm 4.2 A}}$ examine centralizer of a transitive group.

Let G be a transitive subgroup of $Sym(\Omega)$, and α a point in Ω. Let $C = C_{Sym(\Omega)}(G)$. Then:

(i) C is semiregular and $C \simeq N_G(G_\alpha)/G_\alpha$

(ii) C is transitive $\Leftrightarrow$ G is regular.

(iii) If C is transitive, then it is conjugate to G in $Sym(\Omega)$ and hence C is regular

(iv) $C = 1 \Leftrightarrow G_\alpha$ is self-normalizing in G (i.e. $N_G(G_\alpha) = G_\alpha$)

(v) if G is abelian, then $C = G$.

(vi) if G is primitive and nonabelian, then $C = 1$

pf: We apply the preceding lemma with $H = G_\alpha$ and use G and $\rho(G)$ are permutation isomorphic.

(permutation given by $\varphi : G \to \rho(G)$; $g \mapsto \rho(g)$.
$\psi : \Omega \to \Gamma_{G_\alpha}$; $\beta \mapsto G_\alpha g$ where $\alpha^g = \beta$

need transitive $\uparrow$ g

(i) apply lemma (ii) to get C is permutation isomorphic to $\lambda(K)$ where $K = N_G(G_\alpha)$ and $\lambda(K) \simeq K/H = N_G(G_\alpha)/G_\alpha$

(ii) C transitive $\Rightarrow$ $\lambda(K)$ transitive

apply lemma (iv). $K = G$ so $G_\alpha \trianglelefteq G \Rightarrow G_\alpha = G_\beta$ $\forall \beta \in \Omega$

So $G_\alpha = 1$ and G is regular.

Conversely, G regular $\Rightarrow G_\alpha = 1 \Rightarrow G_\alpha \trianglelefteq G \Rightarrow \lambda(K) = G$.

(iii) apply lemma (iv)

(iv) obvious

(v) G abelian $\Rightarrow G_\alpha \trianglelefteq G \Rightarrow G_\alpha = 1 \Rightarrow C = G$.

(vi) $N_G(G_\alpha) = G$ or G_α

if $N_G(G_\alpha) = G \overset{(v)}{\Longrightarrow} C = G$ so G is abelian (by def of C).

So $N_G(G_\alpha) = G_\alpha \Rightarrow C = 1$

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$\boxed{\text{Thm 4.3 B}}$ minimal normal subgroup of a finite primitive group. (primitive : normal → transitive)

If G is a finite primitive subgroup of $Sym(\Omega)$, and K is a minimal normal subgroup of G, then exactly one of the following holds:

(i) for some prime p and some integer d, K is a regular elementary abelian group of order p^d and $soc(G) = K = C_G(K)$

(ii) K is a regular nonabelian group, $C_G(K)$ is a minimal normal subgroup of G which is permutation isomorphic to K and $soc(G) = K \times C_G(K)$.

(iii) K is nonabelian, $C_G(K) = 1$ and $soc(G) = K$.

pf:

Put $C := C_G(K)$. Since $K \triangleleft G$, $C \triangleleft G$ so by thm 1.6A either $C = 1$ or C is transitive. Since K is transitive, by thm 4.2A, $C_{Sym(\Omega)}(K) \supseteq C_G(K)$ is semiregular, so $C_G(K) = C$ is semiregular, hence either $C = 1$ or C is regular.